

Complete Derivations

Every Object and Variable in the Canonical-BSDT Framework

From First Principles to Final Formula

Contents

1	Primitives: State, Features, Metric	7
1.1	State matrix X	7
1.2	Feature map S	7
1.3	Metric G	7
1.4	Derivation of μ_G and M_G for BSDT	8
2	Energy E	8
2.1	Definition	8
2.2	Properties	8
2.3	Time derivative of E	8
2.4	BSDT instantiation of E	8
3	State-Space Gradient g_X	9
3.1	Definition	9
3.2	Derivation from first principles	9
3.3	Matrix form	9
3.4	Norm of g_X	9
3.5	Coercivity bounds on $\ g_X\ $	9
3.6	BSDT pullback gradient $\tilde{G}_t$	10
4	MFLS: Mahalanobis Field Line Score	10
4.1	BSDT definition	10
4.2	Canonical derivation	10
4.3	Expanded forms	10
4.4	MFLS squared	10
4.5	MFLS as Fisher information	10
4.6	MFLS as leading indicator (proved)	11
4.7	Verification against BSDT	11
5	Adaptive Gain γ	11
5.1	Definition	11
5.2	Derivation from the saddle-point game	11
5.3	Properties	12
5.4	Fermi-Dirac structure	12
5.5	Formula for θ	12
6	The Canonical ODE	13
6.1	Locked form	13

6.2	Decomposition	13
6.3	Force decomposition	13
7	Lyapunov Energy Derivative $\dot{E}$	13
7.1	Master identity	13
7.2	Special cases	14
8	Effective Decay Rate ρ_{eff}	14
8.1	Definition	14
8.2	Derivation in canonical terms	14
8.3	Connection to σ	15
8.4	Half-life prediction	15
9	Effective Singular Value σ_{eff}	15
9.1	Abstract definition	15
9.2	Derivation in canonical terms	15
9.3	Properties	15
10	Theoretical Rate ρ	16
10.1	Definition	16
10.2	Derivation	16
10.3	Expression via σ_{eff}	16
11	Admissibility Threshold Ψ^*	16
11.1	Abstract form	16
11.2	Derivation in canonical terms	16
11.3	BSDT instantiation	17
11.4	Sensitivity to calibration conditioning	17
12	Admissibility Condition	17
12.1	Statement	17
12.2	Derivation	17
12.3	The canonical admissibility ratio	18
13	Safety Ratio	18
13.1	Definition	18
13.2	Derivation	18
13.3	Regimes	18
13.4	Connection to effective decay rate	19
14	Alignment Angle $\cos \theta_t$	19
14.1	Definition	19
14.2	Derivation from energy descent	19
14.3	Energy derivative in terms of $\cos \theta_t$	19
14.4	Geometric interpretation	19
14.5	Collapse angle $\tan \theta_t$	19
15	Misalignment Angle ψ_t (Sixth Signal)	20
15.1	Definition	20
15.2	Derivation	20
15.3	What ψ_t measures	20
15.4	False alarm resolution	20
15.5	Performance gap formula	20

16 Channel Recovery Operator $\mathcal{R}$	21
16.1 The recovery formula	21
16.2 Derivation	21
16.3 Per-channel recovery	21
16.4 Attribution	21
16.5 Stability of recovery	21
16.6 Condition number of recovery	21
17 Ultimate Boundedness Threshold Ψ_{rob}^*	22
17.1 Standard case	22
17.2 Robust case with adversarial disturbance	22
17.3 Admissibility condition (robust)	22
18 The Complete Dashboard	22
18.1 All signals from two objects	22
18.2 Dependency graph	23
18.3 Computational algorithm (per step t)	23
19 BSDT–Canonical Juxtaposition	24
19.1 Identity table	24
19.2 The master identity	24
19.3 Performance gap	25
20 Summary Reference	25
20.1 Every formula	25
20.2 The two-object reduction	26
Part II: Per-Channel Derivations	28
21 Calibration Objects	28
21.1 Normal-period statistics	28
22 Channel 1: Camouflage δ_C	28
22.1 Formula	28
22.2 Derivation	28
22.3 Jacobian $\partial\delta_C^{(i)}/\partial x_t^{(i)}$	29
22.4 Full row 1 of J	29
22.5 Geometric interpretation	29
22.6 Statistical interpretation	29
22.7 Domain instantiations	30
23 Channel 2: Feature Gap δ_G	30
23.1 Formula	30
23.2 Derivation	30
23.3 Angle of deviation	30
23.4 Jacobian $\partial\delta_G^{(i)}/\partial x_t^{(i)}$	30
23.5 Full row 2 of J	31
23.6 Why δ_G is the most powerful channel	31
23.7 Relation to δ_C	31
24 Channel 3: Activity Anomaly δ_A	31
24.1 Formula	31

24.2	Derivation	32
24.3	Jacobian $\partial\delta_A^{(i)}/\partial x_t^{(i)}$	32
24.4	Full row 3 of J	32
24.5	Geometric meaning of J_A	32
24.6	Requirement: two consecutive snapshots	33
24.7	Domain instantiations	33
25	Channel 4: Temporal Novelty δ_T	33
25.1	Formula	33
25.2	KDE density estimate	33
25.3	Step-by-step derivation	33
25.4	Kernel weights	34
25.5	Jacobian $\partial\delta_T^{(i)}/\partial x_t^{(i)}$	34
25.6	Full row 4 of J	35
25.7	Interpretation: δ_T vs δ_C	35
25.8	Domain instantiations	35
26	The Assembled Jacobian J	35
26.1	Full matrix	35
26.2	Gram matrix	36
26.3	When channels are orthogonal	36
27	Per-Channel Canonical Gradient	37
27.1	Decomposition	37
27.2	Per-channel gradient vectors	37
27.3	MFLS per channel	37
28	Per-Channel Energy Contributions	38
28.1	Energy decomposition	38
29	Per-Channel Lyapunov Contributions $\dot{V}_k$	38
29.1	Decomposition theorem	38
29.2	Drift and control components	38
29.3	Channel stability attribution	39
29.4	Control efficiency per channel	39
30	Per-Channel Attribution a_k	39
30.1	Energy attribution	39
30.2	MFLS attribution	39
30.3	Dominant channel	40
31	Control Coupling Sum R_t	40
32	Complete Per-Step Algorithm	40
32.1	All objects from raw data	40
33	Complete Formula Reference	42
Part III:	Game-Theoretic Architecture	44
34	The Projection Game	44
34.1	Setup	44

34.2	Saddle-point theorem	44
34.3	Interpretation	45
35	The Adversarial Disturbance Game	45
35.1	Setup	45
35.2	No interior saddle	45
36	The Regularised Adversarial Game	46
36.1	Motivating the regulariser	46
36.2	Regularised game	46
37	The Multi-Engine Cooperative Game	46
37.1	Setup	46
37.2	Required assumption	47
37.3	Conditional superadditivity	47
38	Channel Attribution as Shapley Value	47
38.1	Setup	47
38.2	Shapley axioms	48
38.3	Coalitional value	48
38.4	Shapley attribution theorem	48
38.5	Shapley-to-channel inversion (conditional)	48
39	The θ Mechanism Design Problem	49
39.1	Setup	49
39.2	Admissibility-binding solution	49
40	The Information Game	50
40.1	Setup	50
40.2	Information ordering	50
40.3	Decision-theoretic consequence	50
41	The Channel-Regime Phase Modulator (Heuristic Interpretation)	50
41.1	Honest framing	50
41.2	The phase modulator	51
41.3	Interpretation (not theorem)	51
42	Architecture Summary	51
42.1	What is proved at theorem level	51
42.2	What is NOT claimed	52
42.3	What the architecture says	52
Part IV:	Phase Extension	53
43	Angular Velocity of the Alignment Angle	53
43.1	Exact formula	53
43.2	Derivation of $\dot{h}$	53
43.3	Discrete-time formula	54
43.4	Computable form	54
44	Phase-Amplitude Decomposition of $\dot{E}$	54
44.1	Amplitude-phase split of the state	54

44.2	Velocity split	54
44.3	Energy derivative decomposition	54
44.4	Under the canonical ODE	55
45	Phase Energy and Kuramoto Channel	55
45.1	Extracting phases from canonical states	55
45.2	Phase coherence channel δ_Φ	55
45.3	Kuramoto order parameter	56
45.4	Phase energy	56
45.5	Phase gain	56
46	The Three-Phase Precursor Theorem	56
46.1	The collapse signature	56
46.2	Lead-time structure	57
46.3	Falsifiability	57
47	Phase-Aware Canonical ODE	58
47.1	Motivation	58
47.2	The phase-aware ODE	58
47.3	Energy derivative under phase-aware ODE	58
47.4	Manifold invariance under phase-aware ODE	59
48	Domain Instantiations	59
48.1	Power grid (ERCOT): phase-angle dynamics	59
48.2	EEG (neuroscience): seizure onset	59
48.3	Crypto futures (trading): market cascade	59
48.4	Cardiac fibrillation (speculative)	60
49	Updated Dashboard and Algorithm	60
49.1	New signals	60
49.2	Updated per-step algorithm	60
50	Summary: What Phase Extension Adds	61

1 Primitives: State, Features, Metric

1.1 State matrix X

Definition 1.1 (State). The state is a matrix

$$X \in \mathbb{R}^{N \times d}$$

where N is the number of agents and d is the feature dimension per agent. Row i is the state of agent i : $x_i \in \mathbb{R}^d$. In vectorised form: $\text{vec}(X) \in \mathbb{R}^{Nd}$.

Dimensions.

Symbol	Meaning	Type
N	number of agents	positive integer
d	features per agent	positive integer
$n = Nd$	total state dimension	positive integer
X	full state	$\mathbb{R}^{N \times d}$
x_i	state of agent i	$\mathbb{R}^d$

1.2 Feature map S

Definition 1.2 (Feature map).

$$S : \mathbb{R}^{N \times d} \rightarrow \mathbb{R}^k, \quad X \mapsto S(X)$$

$S(X)$ encodes the geometric content relevant to the control objective. $k \geq 1$ is fixed per domain. The Jacobian of S at X is

$$J(X) = \frac{\partial S}{\partial X} \in \mathbb{R}^{k \times n}, \quad n = Nd.$$

Jacobian in index notation. Let $S = (S^1, \dots, S^k)^\top$ and $X = (X^1, \dots, X^n)^\top$ (vectorised). Then:

$$J_{ab}(X) = \frac{\partial S^a}{\partial X^b}, \quad a \in \{1, \dots, k\}, \quad b \in \{1, \dots, n\}.$$

Chain rule. For any trajectory $X(t)$:

$$\dot{S} = J(X)\dot{X} \in \mathbb{R}^k.$$

1.3 Metric G

Definition 1.3 (Feature-space metric).

$$G \in \mathbb{R}^{k \times k}, \quad G = G^\top \succ 0.$$

Eigenvalue bounds: $0 < \mu_G \leq \lambda(G) \leq M_G$.

Derivation of eigenvalue bounds. Since $G \succ 0$, all eigenvalues are positive. Order them $0 < \lambda_1 \leq \dots \leq \lambda_k$. Define:

$$\mu_G := \lambda_1 = \lambda_{\min}(G), \quad M_G := \lambda_k = \lambda_{\max}(G).$$

For any $v \in \mathbb{R}^k$:

$$\mu_G \|v\|^2 \leq v^\top G v \leq M_G \|v\|^2.$$

These bounds are tight: achieved at the eigenvectors of G .

1.4 Derivation of μ_G and M_G for BSDT

Under the BSDT instantiation $G = \Sigma_0^{-1}$:

$$\mu_G = \lambda_{\min}(\Sigma_0^{-1}) = \frac{1}{\lambda_{\max}(\Sigma_0)}, \quad M_G = \lambda_{\max}(\Sigma_0^{-1}) = \frac{1}{\lambda_{\min}(\Sigma_0)}.$$

Proof: if $\Sigma_0 v = \lambda v$ then $\Sigma_0^{-1} v = \lambda^{-1} v$, so eigenvalues invert. $\square$

Conditioning:

$$\kappa(G) = \frac{M_G}{\mu_G} = \frac{\lambda_{\max}(\Sigma_0)}{\lambda_{\min}(\Sigma_0)} = \kappa(\Sigma_0).$$

2 Energy E

2.1 Definition

$$E(X) = S(X)^\top G S(X) \in \mathbb{R}_{\geq 0}$$

2.2 Properties

Proposition 2.1 (Non-negativity and zero set). $E(X) \geq 0$ with equality if and only if $G^{1/2} S(X) = 0$, i.e. $S(X) = 0$ (since $G \succ 0$).

Proof. $E = S^\top G S = \|G^{1/2} S\|^2 \geq 0$. Equality holds iff $G^{1/2} S = 0$ iff $S = 0$ since $G^{1/2}$ is invertible. $\square$

2.3 Time derivative of E

Proposition 2.2 (Energy velocity). Along any trajectory $X(t)$:

$$\dot{E}(t) = \langle g_X(X(t)), \dot{X}(t) \rangle$$

where $g_X = \nabla_X E$ is derived in Section 3.

Proof. By the chain rule:

$$\dot{E} = \frac{d}{dt}[S^\top G S] = 2S^\top G \dot{S} = 2S^\top G J \dot{X} = (2J^\top G S)^\top \dot{X} = g_X^\top \dot{X}.$$

$\square$

2.4 BSDT instantiation of E

With $S = \tilde{X} \Sigma_0^{-1/2}$ (vectorised), $G = I$:

$$E_{\text{BSDT}} = \left\| \tilde{X} \Sigma_0^{-1/2} \right\|_F^2 = \text{tr}(\tilde{X} \Sigma_0^{-1} \tilde{X}^\top)$$

where $\tilde{X} = X - \mathbf{1} \mu_0^\top$.

With $S = B(X) = (\delta_C, \delta_G, \delta_A, \delta_T)^\top$, $G = A$:

$$E_{\text{BSDT}} = B(X)^\top A B(X) = E_{\text{BS}}(S).$$

3 State-Space Gradient g_X

3.1 Definition

$$g_X(X) = \nabla_X E(X) = 2J(X)^\top G S(X) \in \mathbb{R}^n$$

3.2 Derivation from first principles

Proof. Write $E = \sum_{a,b} G_{ab} S^a S^b$. Differentiate with respect to X^j :

$$\frac{\partial E}{\partial X^j} = \sum_{a,b} G_{ab} \left(\frac{\partial S^a}{\partial X^j} S^b + S^a \frac{\partial S^b}{\partial X^j} \right) = 2 \sum_{a,b} G_{ab} J_{aj} S^b = 2[J^\top G S]_j.$$

Hence $\nabla_X E = 2J^\top G S$. $\square$

3.3 Matrix form

In matrix notation with X as an $N \times d$ matrix (not vectorised), the gradient is an $N \times d$ matrix:

$$\nabla_X E = 2J^\top G S \in \mathbb{R}^{N \times d}$$

where $J \in \mathbb{R}^{k \times Nd}$ acts on $\text{vec}(\dot{X})$ and the result is reshaped.

3.4 Norm of g_X

Proposition 3.1 (Gradient norm).

$$\|g_X\|^2 = 4S^\top G J J^\top G S = 4S^\top G \mathcal{G} G S$$

where $\mathcal{G} = J J^\top \in \mathbb{R}^{k \times k}$ is the Gram matrix of J .

Proof. $\|g_X\|^2 = (2J^\top G S)^\top (2J^\top G S) = 4S^\top G J J^\top G S$. $\square$

3.5 Coercivity bounds on $\|g_X\|$

Proposition 3.2 (Two-sided gradient coercivity). *Under (A1)–(A2) with $\sigma_{\min}(J) \geq \sigma > 0$ and $\|J\|_{\text{op}} \leq J_{\max}$:*

$$C_1 \sqrt{E} \leq \|g_X\| \leq C_2 \sqrt{E}$$

where

$$C_1 = \frac{2\sigma\mu_G}{\sqrt{M_G}}, \quad C_2 = \frac{2J_{\max}M_G}{\sqrt{\mu_G}}.$$

Proof. Lower bound.

$$\|g_X\|^2 = 4 \left\| J^\top G S \right\|^2 \geq 4\sigma_{\min}(J)^2 \|G S\|^2 \geq 4\sigma^2 \mu_G^2 \|S\|^2.$$

Since $E = S^\top G S \leq M_G \|S\|^2$, we have $\|S\|^2 \geq E/M_G$. Therefore $\|g_X\|^2 \geq 4\sigma^2 \mu_G^2 E/M_G = C_1^2 E$.

Upper bound.

$$\|g_X\| = 2 \left\| J^\top G S \right\| \leq 2 \|J\|_{\text{op}} \|G\|_{\text{op}} \|S\| \leq 2J_{\max} M_G \|S\|.$$

Since $E = S^\top G S \geq \mu_G \|S\|^2$, we have $\|S\| \leq \sqrt{E/\mu_G}$. Therefore $\|g_X\| \leq 2J_{\max} M_G \sqrt{E/\mu_G} = C_2 \sqrt{E}$. $\square$

3.6 BSDT pullback gradient $\tilde{G}_t$

In the BSDT system, the gradient decomposes per channel:

$$g_X = 2 \sum_{k \in \{C, G, A, T\}} [AS]_k \frac{\partial \delta_k}{\partial X} =: \tilde{G}_t.$$

This is the pullback gradient of Section XVI.3 of the anatomy document. The canonical formula $g_X = 2J^\top GS$ recovers this exactly under the identification $G = A$, $S = B(X)$, $J = \partial B / \partial X$.

4 MFLS: Mahalanobis Field Line Score

4.1 BSDT definition

$$\text{MFLS}_{\text{BSDT}}(X) = \|\nabla_X E_{\text{BSDT}}\|_F = 2 \left\| \tilde{X} \Sigma_0^{-1} \right\|_F$$

4.2 Canonical derivation

$$\text{MFLS}(X) = \|g_X\| = 2 \left\| J^\top GS \right\|$$

Proof. By definition $\text{MFLS} = \|\nabla_X E\|_F$. The canonical gradient is $\nabla_X E = g_X = 2J^\top GS$. Therefore $\text{MFLS} = \|2J^\top GS\| = \|g_X\|$. $\square$

4.3 Expanded forms

Three equivalent canonical expressions:

Form	Expression	When to use
Gradient norm	$\ g_X\ $	Free — already computed in canonical loop
Explicit	$2 \ J^\top GS\ $	When gradient vector is needed
Gram form	$2\sqrt{S^\top G G S}$	When Gram structure is analysed

4.4 MFLS squared

$$\text{MFLS}^2 = \|g_X\|^2 = 4S^\top G G S = 4S^\top G J J^\top G S$$

4.5 MFLS as Fisher information

Under $G = I_S$ (Fisher metric):

$$\text{MFLS}^2 = 4\text{tr}(\Sigma_0^{-1} \tilde{X}^\top \tilde{X} \Sigma_0^{-1}) = 4\text{tr}(I_{\text{Fisher}})$$

MFLS squared is four times the trace of the Fisher information matrix at the current state. High MFLS means the system is in a region of high log-likelihood curvature.

4.6 MFLS as leading indicator (proved)

From the energy derivative with $F_{\text{base}} = 0$:

$$\dot{E} = -(1 - \gamma) \|g_X\|^2 = -(1 - \gamma) \cdot \text{MFLS}^2.$$

MFLS squared is the instantaneous energy descent rate. A spike in MFLS precedes a drop in E . This is a theorem, not an empirical observation.

4.7 Verification against BSDT

Under $S = B(X)$, $G = \Sigma_0^{-1}$, $J = \partial B / \partial X$ with $J_C = 2\tilde{X}\Sigma_0^{-1}$ (the camouflage Jacobian):

$$g_X = 2J_C^T \Sigma_0^{-1} S = 2\tilde{X}\Sigma_0^{-1}, \quad \|g_X\|_F = 2 \left\| \tilde{X}\Sigma_0^{-1} \right\|_F = \text{MFLS}_{\text{BSDT}}. \checkmark$$

5 Adaptive Gain γ

5.1 Definition

$$\gamma(X) = \frac{E(X)}{E(X) + \theta}, \quad \theta > 0$$

5.2 Derivation from the saddle-point game

Consider the two-player zero-sum game at fixed X :

- Controller chooses $\gamma \in [0, 1)$
- Adversary chooses $u \in \mathbb{R}^n$ with $\|u\| \leq M_{\text{adv}}$
- Payoff to adversary: $J(\gamma, u) = \dot{E}(X; \gamma, u)$

The payoff under the canonical ODE is:

$$J(\gamma, u) = (1 - \gamma) \left[\langle g_X, F_{\text{base}} + u \rangle - \|g_X\|^2 \right].$$

The adversary maximises: by Cauchy–Schwarz, $u^* = M_{\text{adv}} g_X / \|g_X\|$.

The controller minimises:

$$\inf_{\gamma \in [0, 1)} (1 - \gamma) B, \quad B = \langle g_X, F_{\text{base}} \rangle + M_{\text{adv}} \|g_X\| - \|g_X\|^2.$$

When $B > 0$: infimum is 0 as $\gamma \rightarrow 1^-$; boundary solution. When $B \leq 0$: minimum at $\gamma = 0$.

The Lagrangian formulation (Theorem O.1 of v4):

$$\mathcal{L}(u, \lambda) = \frac{1}{2} \|u - F_{\text{base}}\|^2 + \lambda \langle u, g_X \rangle - \frac{\theta \|g_X\|^2}{2E} \lambda^2$$

has unique saddle point with $\lambda^* = \frac{\langle F_{\text{base}}, g_X, E \rangle}{(E + \theta) \|g_X\|^2}$, giving $\gamma = E / (E + \theta)$.

5.3 Properties

Proposition 5.1 (Gain properties). (i) $\gamma \in [0, 1)$ for all X

(ii) $\gamma \rightarrow 0$ as $E \rightarrow 0$: no damping near target

(iii) $\gamma \rightarrow 1$ as $E \rightarrow \infty$: full projection at high energy

(iv) $\gamma = 1/2$ when $E = \theta$: maximum-entropy operating point

(v) $\gamma' = \theta/(E + \theta)^2 > 0$: strictly increasing in E

Proof. (i) $E \geq 0, \theta > 0$, so $0 \leq E/(E + \theta) < 1$. (ii) $\lim_{E \rightarrow 0} E/(E + \theta) = 0$. (iii) $\lim_{E \rightarrow \infty} E/(E + \theta) = 1$. (iv) $E/(E + \theta) = 1/2 \iff E = \theta$. (v) $d\gamma/dE = \theta/(E + \theta)^2 > 0$. $\square$

5.4 Fermi-Dirac structure

Writing $\gamma = 1/(1 + \theta/E)$: this is the Fermi-Dirac occupation function with inverse temperature $\beta = 1/\theta$ and energy E . The thermodynamic dictionary (v4 §N):

Canonical object	Thermodynamic analogue
θ	Temperature
γ	Occupation number
$1 - \gamma$	Carnot efficiency
$E = \theta$	Maximum-entropy operating point
$\theta \rightarrow 0$	Zero temperature (full intervention)
$\theta \rightarrow \infty$	Infinite temperature (no intervention)

5.5 Formula for θ

$$\theta = \text{median}_{t \in \text{calm}} \{E(t)\} = \text{median}_{t \in \text{calm}} \left\{ \left\| \tilde{X}_t \Sigma_0^{-1/2} \right\|_F^2 \right\}$$

Derivation. The median of the calm-period energy distribution is the natural temperature because:

1. At $E = \theta, \gamma = 1/2$ — half intervention. During normal operation ($E \approx \theta$ by construction), the system applies moderate damping.
2. The median is robust to outlier spikes in the calibration window. The mean would overstate θ .
3. Setting $\theta = E_{\text{median}}$ means the system is at its maximum-entropy operating point on average during the calm regime — it neither over-intervenes nor under-intervenes.

Alternative derivations of θ :

Route 1 (Maximum entropy): set $\theta = e^*$ (critical manifold energy) so $\gamma = 1/2$ at stability boundary.

Route 2 (Admissibility): choose $\theta > M_{\max} \sqrt{M_G}/(\sigma \mu_G)$ to guarantee $\Psi^* > M_{\max}$.

Route 3 (Chi-squared calibration): $e^* = \chi_{Nd, 0.99}^2$, set $\theta = e^*/2$ so $\gamma = 1/3$ at the 99% energy quantile.

6 The Canonical ODE

6.1 Locked form

$$\dot{X} = F_{\text{base}}(X) - g_X(X) - \gamma(X) \frac{\langle F_{\text{base}}(X) - g_X(X), g_X(X) \rangle}{\|g_X(X)\|^2} g_X(X)$$

6.2 Decomposition

Define the modified force:

$$F(X) = F_{\text{base}}(X) - g_X(X).$$

The projection coefficient:

$$\alpha(X) = \frac{\langle F(X), g_X(X) \rangle}{\|g_X(X)\|^2}.$$

The ODE becomes:

$$\dot{X} = F(X) - \gamma(X)\alpha(X)g_X(X).$$

Three-term decomposition:

1. $F_{\text{base}}(X)$: domain-specific nominal field
2. $-g_X(X)$: gradient correction forcing E to decrease
3. $-\gamma\alpha g_X$: adaptive projection removing the component of F aligned with g_X

6.3 Force decomposition

$$\dot{X} = \underbrace{F - \langle F, \hat{u} \rangle \hat{u}}_{F_{\perp} \text{ (always free)}} + \underbrace{(1 - \gamma) \langle F, \hat{u} \rangle \hat{u}}_{(1-\gamma)F_{\parallel} \text{ (damped)}}$$

where $\hat{u} = g_X / \|g_X\|$.

At $E \rightarrow 0$: $\gamma \rightarrow 0$, $\dot{X} \rightarrow F_{\text{base}}$ (free dynamics). At $E \rightarrow \infty$: $\gamma \rightarrow 1$, $\dot{X} \rightarrow F_{\perp}$ (only safe motion).

7 Lyapunov Energy Derivative $\dot{E}$

7.1 Master identity

Theorem 7.1 (Energy derivative, Lemma 7.7 of v4). *Along any solution of the canonical ODE:*

$$\dot{E} = (1 - \gamma) \left[\langle g_X, F_{\text{base}} \rangle - \|g_X\|^2 \right]$$

Proof.

$$\begin{aligned}
\dot{E} &= \langle g_X, \dot{X} \rangle \\
&= \langle g_X, F_{\text{base}} - g_X - \gamma \alpha g_X \rangle \\
&= \langle g_X, F_{\text{base}} \rangle - \|g_X\|^2 - \gamma \alpha \|g_X\|^2 \\
&= \langle g_X, F_{\text{base}} \rangle - \|g_X\|^2 - \gamma \langle F, g_X \rangle \\
&= \langle g_X, F_{\text{base}} \rangle - \|g_X\|^2 - \gamma \langle F_{\text{base}} - g_X, g_X \rangle \\
&= \langle g_X, F_{\text{base}} \rangle - \|g_X\|^2 - \gamma \langle F_{\text{base}}, g_X \rangle + \gamma \|g_X\|^2 \\
&= (1 - \gamma) \langle g_X, F_{\text{base}} \rangle - (1 - \gamma) \|g_X\|^2 \\
&= (1 - \gamma) \left[\langle g_X, F_{\text{base}} \rangle - \|g_X\|^2 \right]. \quad \square
\end{aligned}$$

□

7.2 Special cases

Pure geometric flow ($F_{\text{base}} = 0$):

$$\dot{E} = -(1 - \gamma) \|g_X\|^2 = -(1 - \gamma) \cdot \text{MFLS}^2 \leq 0.$$

Strictly negative whenever $g_X \neq 0$ and $\gamma < 1$.

Descent condition (general F_{base}):

$$\dot{E} \leq 0 \iff \langle g_X, F_{\text{base}} \rangle \leq \|g_X\|^2.$$

At the maximum-entropy point ($E = \theta$, $\gamma = 1/2$):

$$\dot{E} = \frac{1}{2} \left[\langle g_X, F_{\text{base}} \rangle - \|g_X\|^2 \right].$$

8 Effective Decay Rate ρ_{eff}

8.1 Definition

$$\rho_{\text{eff}}(t) = -\frac{\dot{E}(t)}{E(t)}$$

8.2 Derivation in canonical terms

From the energy derivative:

$$\rho_{\text{eff}} = \frac{(1 - \gamma) \left[\|g_X\|^2 - \langle g_X, F_{\text{base}} \rangle \right]}{E}.$$

Under $F_{\text{base}} = 0$:

$$\rho_{\text{eff}} = \frac{(1 - \gamma) \|g_X\|^2}{E} = \frac{(1 - \gamma) \cdot \text{MFLS}^2}{E}.$$

8.3 Connection to σ

From the coercivity bound $\|g_X\|^2 \geq C_1^2 E$:

$$\rho_{\text{eff}} \geq (1 - \gamma)C_1^2 = (1 - \gamma)\frac{4\sigma^2\mu_G^2}{M_G} = \rho_{\text{theoretical}}.$$

The theoretical rate ρ is the worst-case lower bound. The effective rate ρ_{eff} is the instantaneous observed rate. The ratio $\rho_{\text{eff}}/\rho \geq 1$ measures how much better the system is performing relative to the theoretical guarantee.

8.4 Half-life prediction

The local exponential half-life is:

$$\tau_{1/2}(t) = \frac{\ln 2}{\rho_{\text{eff}}(t)}.$$

This is the time at which E will reach $E(t)/2$ at the current decay rate.

9 Effective Singular Value σ_{eff}

9.1 Abstract definition

$\sigma = \sigma_{\min}(J(X))$ from Theorem 9.3 of v4 is the abstract worst-case bound.

9.2 Derivation in canonical terms

$$\sigma_{\text{eff}}(t) = \frac{\|g_X(t)\| \sqrt{M_G}}{2\mu_G \sqrt{E(t)}} = \frac{\text{MFLS}_t \sqrt{M_G}}{4\mu_G \sqrt{E_t}}$$

Proof. From the coercivity bound:

$$\|g_X\|^2 \geq \frac{4\sigma^2\mu_G^2}{M_G} E.$$

Treat this as an equality to define the effective σ that is tight at the current state:

$$\|g_X\|^2 = \frac{4\sigma_{\text{eff}}^2\mu_G^2}{M_G} E \implies \sigma_{\text{eff}}^2 = \frac{\|g_X\|^2 M_G}{4\mu_G^2 E} \implies \sigma_{\text{eff}} = \frac{\|g_X\| \sqrt{M_G}}{2\mu_G \sqrt{E}}.$$

Since $\text{MFLS} = \|g_X\|$ (Section 4):

$$\sigma_{\text{eff}} = \frac{\text{MFLS} \sqrt{M_G}}{2\mu_G \sqrt{E}}.$$

□

9.3 Properties

$\sigma_{\text{eff}}(t) \geq \sigma$ always (effective is at least as good as worst case).

At the coercivity lower bound: $\sigma_{\text{eff}} = \sigma$ (bound is tight when J is at its worst orientation).

σ_{eff} is a time-varying signal measuring the current Jacobian rank quality — computable at each step.

10 Theoretical Rate ρ

10.1 Definition

$$\rho = \frac{4\sigma^2\mu_G^2}{M_G}(1 - \gamma_{\max}), \quad \gamma_{\max} = \frac{E(X_0)}{E(X_0) + \theta}$$

10.2 Derivation

From Theorem 9.3 of v4 with $F_{\text{base}} = 0$:

$$\dot{E} = -(1 - \gamma) \|g_X\|^2.$$

Apply the coercivity lower bound:

$$\|g_X\|^2 \geq \frac{4\sigma^2\mu_G^2}{M_G}E.$$

Since $\gamma \leq \gamma_{\max} = E_0/(E_0 + \theta) < 1$ (monotone in E , decreasing, so maximum at $t = 0$):

$$\dot{E} \leq -(1 - \gamma_{\max}) \frac{4\sigma^2\mu_G^2}{M_G}E = -\rho E.$$

By Grönwall: $E(t) \leq E_0 e^{-\rho t}$.

10.3 Expression via σ_{eff}

Substituting $\sigma = \sigma_{\text{eff}}$:

$$\rho_{\text{eff}} = \frac{4\sigma_{\text{eff}}^2\mu_G^2}{M_G}(1 - \gamma) = \frac{4 \cdot \frac{\|g_X\|^2 M_G}{4\mu_G^2 E} \cdot \mu_G^2}{M_G}(1 - \gamma) = \frac{\|g_X\|^2}{E}(1 - \gamma). \checkmark$$

Consistent with the direct definition of ρ_{eff} .

11 Admissibility Threshold Ψ^*

11.1 Abstract form

From Theorem 9.4 of v4:

$$\Psi^* = \frac{\sigma\theta\mu_G}{\sqrt{M_G}}.$$

This is unimodal with maximum at $R^* = \theta$.

11.2 Derivation in canonical terms

$$\Psi^* = \frac{\theta \|g_X\|}{4\sqrt{E}} = \frac{\theta \cdot \text{MFLS}}{4\sqrt{E}}$$

Proof. Substitute $\sigma = \sigma_{\text{eff}} = \|g_X\| \sqrt{M_G}/(2\mu_G\sqrt{E})$:

$$\Psi^* = \frac{\sigma_{\text{eff}}\theta\mu_G}{\sqrt{M_G}} = \frac{\|g_X\| \sqrt{M_G}}{2\mu_G\sqrt{E}} \cdot \frac{\theta\mu_G}{\sqrt{M_G}} = \frac{\theta \|g_X\|}{2\sqrt{E}}.$$

Since $\text{MFLS} = \|g_X\|$:

$$\Psi^* = \frac{\theta \cdot \text{MFLS}}{2\sqrt{E}}.$$

□

Note on the factor. The formula above uses σ_{eff} (instantaneous effective singular value). Using the abstract worst-case σ gives $\Psi_{\text{abstract}}^* = \sigma\theta\mu_G/\sqrt{M_G}$, while the canonical instantaneous formula gives $\Psi^*(t) = \theta\text{MFLS}_t/(2\sqrt{E_t})$. Both are valid; the canonical form is computable at each step.

11.3 BSDT instantiation

With $G = \Sigma_0^{-1}$, $\mu_G = 1/\lambda_{\max}(\Sigma_0)$, $M_G = 1/\lambda_{\min}(\Sigma_0)$:

$$\Psi_{\text{BSDT}}^* = \frac{\sigma\theta/\lambda_{\max}(\Sigma_0)}{\sqrt{1/\lambda_{\min}(\Sigma_0)}} = \frac{\sigma\theta\sqrt{\lambda_{\min}(\Sigma_0)}}{\lambda_{\max}(\Sigma_0)} = \frac{\sigma\theta}{\sqrt{\kappa(\Sigma_0)} \cdot \lambda_{\max}^{1/2}(\Sigma_0)}.$$

The canonical form $\theta\text{MFLS}/(2\sqrt{E})$ is metric-free — the Σ_0 terms cancel. □

11.4 Sensitivity to calibration conditioning

From the BSDT formula:

$$\Psi_{\text{BSDT}}^* \propto \frac{1}{\sqrt{\kappa(\Sigma_0)}}.$$

A $10\times$ increase in calibration conditioning shrinks Ψ^* by $\sqrt{10} \approx 3.16\times$. This is the gap closure result G7.2.

12 Admissibility Condition

12.1 Statement

$$\Psi^* > M_{\max} \iff \frac{\theta \cdot \text{MFLS}_t}{2\sqrt{E_t}} > M_{\max} \iff \frac{\text{MFLS}_t}{\sqrt{E_t}} > \frac{2M_{\max}}{\theta}$$

12.2 Derivation

Starting from the abstract condition:

$$\Psi^* > M_{\max}$$

Substituting the canonical form:

$$\frac{\theta \cdot \text{MFLS}}{2\sqrt{E}} > M_{\max}$$

Rearranging:

$$\text{MFLS} > \frac{2M_{\max}\sqrt{E}}{\theta}$$

Dividing both sides by $\sqrt{E}$:

$$\frac{\text{MFLS}}{\sqrt{E}} > \frac{2M_{\max}}{\theta}.$$

12.3 The canonical admissibility ratio

$$\mathcal{A}(t) = \frac{\text{MFLS}_t}{\sqrt{E_t}} = \frac{\|g_X(t)\|}{\sqrt{E(t)}}$$

This ratio is the canonical admissibility signal. It measures gradient strength relative to energy level.

The admissibility condition is then simply:

$$\mathcal{A}(t) > \frac{2M_{\max}}{\theta}.$$

Connection to ρ_{eff} :

$$\rho_{\text{eff}}(t) = \mathcal{A}(t)^2(1 - \gamma(t)).$$

A system decaying fast is a system that is safe. Fast decay implies admissibility.

13 Safety Ratio

13.1 Definition

$$\text{SafetyRatio}(t) = \frac{\Psi^*(t)}{M_{\max}} = \frac{\theta \cdot \text{MFLS}_t}{2M_{\max}\sqrt{E_t}}$$

13.2 Derivation

Divide the admissibility condition by $M_{\max}$:

$$\text{SafetyRatio} = \frac{\Psi^*}{M_{\max}} = \frac{\theta \text{MFLS}}{2M_{\max}\sqrt{E}}.$$

Substituting $\text{MFLS} = \|g_X\|$, E , θ — all already computed in the canonical loop:

Zero additional computation.

13.3 Regimes

SafetyRatio	Regime	Interpretation
> 1	Admissible	$\Psi^* > M_{\max}$; ultimate boundedness guaranteed
$= 1$	Boundary	Exactly at the admissibility threshold
< 1	Inadmissible	Disturbance can overcome canonical brake
$\rightarrow 0$	Critical	$\text{MFLS} \rightarrow 0$ or $E \rightarrow \infty$; system losing gradient signal

13.4 Connection to effective decay rate

$$\text{SafetyRatio}(t) = \frac{\theta}{2M_{\max}} \mathcal{A}(t) = \frac{\theta}{2M_{\max}} \sqrt{\frac{\rho_{\text{eff}}(t)}{1 - \gamma(t)}}.$$

This connects the safety guarantee directly to the observable performance metric.

14 Alignment Angle $\cos \theta_t$

14.1 Definition

$$\cos \theta_t = \frac{\langle F_{\text{base}}(X_t), g_X(X_t) \rangle}{\|F_{\text{base}}(X_t)\| \cdot \|g_X(X_t)\|} \in [-1, +1]$$

14.2 Derivation from energy descent

The descent condition $\dot{E} \leq 0$ requires:

$$\langle g_X, F_{\text{base}} \rangle \leq \|g_X\|^2.$$

Writing $\langle g_X, F_{\text{base}} \rangle = \|g_X\| \|F_{\text{base}}\| \cos \theta_t$:

$$\|F_{\text{base}}\| \cos \theta_t \leq \|g_X\|.$$

When $\cos \theta_t \leq 0$: descent is automatic regardless of $\|F_{\text{base}}\|$. When $\cos \theta_t > 0$: descent requires $\|F_{\text{base}}\| \cos \theta_t < \|g_X\|$, i.e. F_{base} is not pushing into the energy gradient too strongly.

14.3 Energy derivative in terms of $\cos \theta_t$

$$\dot{E} = (1 - \gamma) \|g_X\| [\|F_{\text{base}}\| \cos \theta_t - \|g_X\|].$$

14.4 Geometric interpretation

Value	Meaning
$\cos \theta_t \rightarrow +1$	F_{base} co-aligned with g_X — coherent collapse pressure
$\cos \theta_t \approx 0$	F_{base} tangent to energy level set — no radial progress
$\cos \theta_t \rightarrow -1$	Anti-aligned — restoring competition, E falls without F_{base} help

14.5 Collapse angle $\tan \theta_t$

From §XIV of the anatomy document:

$$\tan \theta_t = \frac{\|F_{\perp}\|}{\|F_{\parallel}\|} = \sqrt{\frac{1}{\cos^2 \theta_t} - 1} = \frac{\sqrt{1 - \cos^2 \theta_t}}{\cos \theta_t}.$$

$\tan \theta_t \rightarrow 0$: trajectory fully collapse-directed. $\tan \theta_t \rightarrow \infty$: motion fully perpendicular, locally safe. $\tan \theta_t = 1$ ($\theta_t = 45$): critical balance.

15 Misalignment Angle ψ_t (Sixth Signal)

15.1 Definition

$$\cos \psi_t = \frac{\langle \tilde{G}_t, g_t \rangle_F}{\|\tilde{G}_t\|_F \cdot \|g_t\|}$$

where $g_t = \nabla_S E_{BS}(S_t) \in \mathbb{R}^4$ is the channel-space gradient and $\tilde{G}_t = J_t^\top g_t \in \mathbb{R}^{Nd}$ is its state-space pullback.

15.2 Derivation

The BSDT master operator uses g_t (channel space) in the projection. The canonical ODE uses $\tilde{G}_t$ (state-space pullback).

These differ by the Jacobian map: $\tilde{G}_t = J_t^\top g_t$.

The angle between them:

$$\cos \psi_t = \frac{\langle \tilde{G}_t, g_t \rangle_F}{\|\tilde{G}_t\|_F \|g_t\|} = \frac{R_t}{\|\tilde{G}_t\|_F \|g_t\|}$$

where $R_t = \sum_k [g_t]_k \langle \partial_X \delta_k, g_t \rangle_F$ is the control coupling sum.

15.3 What ψ_t measures

$\psi_t \approx 0$: channel-space risk and physical dynamics pointing in the same direction. Collapse is real. All five prior signals are reliable.

$\psi_t \approx \pi/2$: complete misalignment. Abstract risk elevated but physical dynamics not aligned with collapse. This is a false alarm.

15.4 False alarm resolution

Previously: high MFLS but no actual collapse. Unexplained by any single-space signal.

Now: high channel MFLS ($\|g_t\|$ large) with ψ_t near $\pi/2$ means $\tilde{G}_t \perp g_t$ — physical dynamics are tangential to the collapse direction. The alarm is real in channel space but not in state space. ψ_t discriminates between the two.

15.5 Performance gap formula

The performance difference between BSDT and CEK-v4:

$$\Delta \text{performance}(t) \approx f(\psi_t) + \kappa(\mathcal{R}_t)^{-1}$$

where:

- ψ_t large: BSDT projects using wrong direction, canonical corrects
- $\kappa(\mathcal{R}_t)$ large: channels nearly collinear, canonical compression is more robust

16 Channel Recovery Operator $\mathcal{R}$

16.1 The recovery formula

$$S_t = \mathcal{R}(X_t)g_X(t), \quad \mathcal{R}(X) = \frac{1}{2}A^{-1}(JJ^\top)^{-1}J \in \mathbb{R}^{k \times n}$$

16.2 Derivation

From $g_X = 2J^\top GS$, left-multiply by J :

$$Jg_X = 2JJ^\top GS.$$

Since $JJ^\top \in \mathbb{R}^{k \times k}$ is invertible under the rank condition:

$$GS = \frac{1}{2}(JJ^\top)^{-1}Jg_X.$$

Since $G = A \succ 0$ is invertible:

$$S = \frac{1}{2}G^{-1}(JJ^\top)^{-1}Jg_X = \frac{1}{2}A^{-1}(JJ^\top)^{-1}Jg_X = \mathcal{R}g_X.$$

16.3 Per-channel recovery

$$[\delta_k] = [S]_k = [\mathcal{R}g_X]_k, \quad k \in \{C, G, A, T\}.$$

16.4 Attribution

$$a_k(t) = \frac{\tilde{S}_k(t)^2}{\|\tilde{S}_t\|^2}, \quad \tilde{S}_k = \frac{S_k}{\mu_k^{\text{normal}}}, \quad \sum_k a_k = 1.$$

16.5 Stability of recovery

Proposition 16.1 (Lipschitz stability). *For perturbed gradient $\tilde{g}_X = g_X + \varepsilon$ with $\|\varepsilon\| \leq \delta$:*

$$\|\tilde{S} - S\| \leq \|\mathcal{R}\|_{\text{op}} \cdot \delta = \frac{\delta}{2\sigma_{\min}(A) \cdot \sigma_{\min}(JJ^\top)^{1/2} \cdot \sigma_{\min}(J)}.$$

Proof. $\tilde{S} - S = \mathcal{R}\varepsilon$, so $\|\tilde{S} - S\| \leq \|\mathcal{R}\|_{\text{op}} \|\varepsilon\|$. Expanding $\|\mathcal{R}\|_{\text{op}} \leq \frac{1}{2} \|A^{-1}\|_{\text{op}} \|(JJ^\top)^{-1}\|_{\text{op}} \|J\|_{\text{op}}$ and substituting singular value bounds gives the result. $\square$

16.6 Condition number of recovery

$$\kappa(\mathcal{R}) = \frac{\sigma_{\max}(\mathcal{R})}{\sigma_{\min}(\mathcal{R})}.$$

Recovery is well-conditioned when $\kappa(\mathcal{R}) \ll 1$.

Recovery fails (ill-conditioned) when:

- Channels nearly collinear: $\sigma_{\min}(JJ^\top) \approx 0$
- Coupling matrix ill-conditioned: $\kappa(A) \gg 1$
- Jacobian rank-deficient: $\sigma_{\min}(J) \approx 0$ (channel has zero sensitivity)

17 Ultimate Boundedness Threshold Ψ_{rob}^*

17.1 Standard case

From Theorem 9.4 of v4:

$$\Psi_{\text{std}}(R) = (1 - \gamma(R)) \cdot 2\sigma \sqrt{\frac{\mu_G^2}{M_G}} R = \frac{\theta}{R + \theta} \cdot C_1 \sqrt{R}.$$

Setting $u = \sqrt{R}$:

$$\Psi_{\text{std}}(u) = \frac{\theta C_1 u}{u^2 + \theta}.$$

Maximum at $u = \sqrt{\theta}$ (i.e. $R = \theta$):

$$\Psi_{\text{std}}^* = \frac{C_1 \sqrt{\theta}}{2}.$$

17.2 Robust case with adversarial disturbance

With two-sided coercivity $C_1 \sqrt{E} \leq \|g_X\| \leq C_2 \sqrt{E}$:

$$\Psi_{\text{rob}}(R) = \frac{\theta C_1 \sqrt{R}}{R + M_{\text{adv}} C_2 \sqrt{R} + \theta} - M_{\text{adv}}.$$

Setting $u = \sqrt{R}$:

$$\frac{d\Psi_{\text{rob}}}{du} = \frac{\theta C_1 (\theta - u^2)}{(u^2 + M_{\text{adv}} C_2 u + \theta)^2}.$$

Positive for $u < \sqrt{\theta}$, negative for $u > \sqrt{\theta}$: unique maximum at $u = \sqrt{\theta}$, i.e. $R = \theta$.

Unimodality proved.

17.3 Admissibility condition (robust)

$$M_{\text{nom}} < \Psi_{\text{rob}}^* = \Psi_{\text{rob}}(\theta) = \frac{\theta C_1 \sqrt{\theta}}{\theta + M_{\text{adv}} C_2 \sqrt{\theta} + \theta} - M_{\text{adv}}.$$

18 The Complete Dashboard

18.1 All signals from two objects

Every signal in the canonical dashboard is a function of g_X and E :

Signal	Formula	Objects	Section
MFLS $_t$	$\ g_X\ $	g_X	§4
E_t	$S^\top G S$	S, G	§2
γ_t	$E/(E + \theta)$	E, θ	§5
$\dot{E}_t$	$(1 - \gamma)(\langle g_X, F_{\text{base}} \rangle - \ g_X\ ^2)$	$g_X, E, \theta, F_{\text{base}}$	§7
$\rho_{\text{eff}}(t)$	$\ g_X\ ^2 (1 - \gamma)/E$	g_X, E, θ	§8
$\cos \theta_t$	$\langle F_{\text{base}}, g_X, / \rangle (\ F_{\text{base}}\ \ g_X\)$	g_X, F_{base}	§11
$\tan \theta_t$	$\sqrt{1/\cos^2 \theta_t - 1}$	$\cos \theta_t$	§11
$\sigma_{\text{eff}}(t)$	$\ g_X\ \sqrt{M_G}/(2\mu_G\sqrt{E})$	g_X, E, μ_G, M_G	§9
$\Psi^*(t)$	$\theta \ g_X\ /(2\sqrt{E})$	g_X, E, θ	§10
$\mathcal{A}(t)$	$\ g_X\ /\sqrt{E}$	g_X, E	§10
SafetyRatio(t)	$\theta \ g_X\ /(2M_{\text{max}}\sqrt{E})$	g_X, E, θ	§10
$\tau_{1/2}(t)$	$\ln 2/\rho_{\text{eff}}(t)$	ρ_{eff}	§8
ψ_t	$\arccos(R_t/(\ \tilde{G}_t\ \ g_t\))$	J, g_t, g_X	§12
S_t (channels)	$\mathcal{R}(X_t)g_X(t)$	J, A, g_X	§13
$a_k(t)$	$\tilde{S}_k^2/\ \tilde{S}\ ^2$	S_t	§13

18.2 Dependency graph

```

X
+-- S = B(X)          --> E = S^T G S
|
|                      |
|                      +-- gamma = E/(E+theta)
|                      +-- gamma_max = E_0/(E_0+theta)
|                      +-- rho = 4sigma^2 mu_G^2/M_G (1-gamma_max)
|
+-- J = dS/dX          --> g_X = 2J^T G S    (= MFLS direction)
|
|                      +-- MFLS = ||g_X||
|                      +-- sigma_eff = MFLS sqrt(M_G)/(2 mu_G sqrt(E))
|                      +-- Psi* = theta MFLS/(2 sqrt(E))
|                      +-- SafetyRatio = Psi*/M_max
|                      +-- rho_eff = MFLS^2 (1-gamma)/E
|                      +-- cos(theta_t) = <F_base,g_X>/(||F_base|| MFLS)
|                      +-- R(X_t) g_X(t) --> channels S_t --> a_k(t)

```

18.3 Computational algorithm (per step t)

1. Compute $\tilde{X}_t = X_t - \mathbf{1}\mu_0^\top$
2. Compute $S_t = B(X_t) = (\delta_C, \delta_G, \delta_A, \delta_T)^\top$
3. Compute $J_t = \partial B/\partial X|_{X_t}$

4. Compute $g_X(t) = 2J_t^\top GS_t$ (canonical gradient)
5. Compute $E_t = S_t^\top GS_t$ (energy)
6. Compute $\gamma_t = E_t/(E_t + \theta)$ (gain)
7. Compute $\text{MFLS}_t = \|g_X(t)\|$ (free from step 4)
8. Compute $\rho_{\text{eff}}(t) = \text{MFLS}_t^2(1 - \gamma_t)/E_t$
9. Compute $\Psi^*(t) = \theta \text{MFLS}_t/(2\sqrt{E_t})$
10. Compute $\text{SafetyRatio}(t) = \Psi^*(t)/M_{\text{max}}$
11. Compute $\cos \theta_t = \langle F_{\text{base}}(X_t), g_X(t) \rangle / (\|F_{\text{base}}\| \text{MFLS}_t)$
12. Compute $\mathcal{R}_t = \frac{1}{2}A^{-1}(J_t J_t^\top)^{-1}J_t$ (4×4 inversion)
13. Compute $S_t = \mathcal{R}_t g_X(t)$ (channel recovery)
14. Compute $a_k(t) = \tilde{S}_k(t)^2 / \|\tilde{S}_t\|^2$ (attribution)
15. Compute $\psi_t = \arccos(R_t / (\|\tilde{G}_t\|_F \|g_t\|))$
16. Update ODE: $\dot{X} = F_{\text{base}} - g_X - \gamma_t \alpha_t g_X$

19 BSDT–Canonical Juxtaposition

19.1 Identity table

Object	BSDT form	Canonical form	Equal?
State	$X \in \mathbb{R}^{N \times d}$	$X \in \mathbb{R}^{N \times d}$	Identical
Energy	$e_t = \left\ \tilde{X} \Sigma_0^{-1/2} \right\ _F^2$	$E = S^\top GS$	Equal under $G = A$
Gradient	$\tilde{G}_t = \sum_k [AS]_k \partial_X \delta_k$	$g_X = 2J^\top GS$	Identical
Gain	$\gamma^* = e_t/(e_t + \theta)$	$\gamma = E/(E + \theta)$	Identical
ODE	$\dot{X} = F_t - \gamma^* \frac{\langle F_t, g_t, \rangle}{\ g_t\ ^2} g_t$	locked canonical ODE	Identical
MFLS	$2 \left\ \tilde{X} \Sigma_0^{-1} \right\ _F$	$\ g_X\ $	Identical
Rate ρ	not proved	$4\sigma^2 \mu_G^2 / M_G(1 - \gamma_{\text{max}})$	canonical proves BSDT
Ψ^*	not proved	$\theta \text{MFLS} / (2\sqrt{E})$	canonical proves BSDT
Channels	$(\delta_C, \delta_G, \delta_A, \delta_T)$ explicit	recovered via $\mathcal{R}g_X$	4×4 inversion
ψ_t	not in BSDT	$\arccos(R_t / \ \tilde{G}_t\ \ g_t\)$	new in canonical
Extensions	none	7 extensions	canonical adds

19.2 The master identity

$$\underbrace{S_t}_{\text{BSDT channels}} = \underbrace{\frac{1}{2}A^{-1}(J_t J_t^\top)^{-1}J_t}_{\text{recovery operator } \mathcal{R}(X_t)} \underbrace{g_X(t)}_{\text{canonical gradient}}$$

BSDT is on the left. Canonical is on the right. The recovery operator in the middle is the bridge. The equation says they are the same object, read from different directions.

19.3 Performance gap

BSDT uses g_t (channel space) in the projection. Canonical uses $\tilde{G}_t$ (state-space pullback). The gap is ψ_t :

$$\Delta\text{performance}(t) \sim \psi_t + \kappa(\mathcal{R}_t)^{-1}.$$

The v34→v58 empirical progression was systematically reducing ψ_t and $\kappa(\mathcal{R}_t)$ without knowing it:

Version	Implicit improvement	Canonical equivalent
v34	Base	No channel recovery
v48	4-quadrant modulator	Partial recovery via $(G_{\text{mem}}, T_{\text{mem}})$
v52	Clip structure	Excludes high $\kappa(\mathcal{R})$ regimes
v55	State-dependent $\theta_G(t)$	Ricci-flow metric adaptation
v58	Asymmetric clamp	Ill-conditioned recovery region excluded

20 Summary Reference

20.1 Every formula

$S = B(X) = (\delta_C, \delta_G, \delta_A, \delta_T)^\top$	feature vector
$E = S^\top GS$	Lyapunov energy
$J = \partial S / \partial X \in \mathbb{R}^{k \times n}$	Jacobian
$g_X = 2J^\top GS$	canonical gradient
$\text{MFLS} = \ g_X\ = 2 \left\ J^\top GS \right\ $	field line score
$\gamma = \frac{E}{E + \theta}$	adaptive gain
$\theta = \text{median}_{t \in \text{calm}} \{E(t)\}$	intervention cost
$\dot{E} = (1 - \gamma) \left[\langle g_X, F_{\text{base}} \rangle - \ g_X\ ^2 \right]$	energy derivative
$\rho_{\text{eff}} = \frac{\ g_X\ ^2 (1 - \gamma)}{E} = \frac{\text{MFLS}^2 (1 - \gamma)}{E}$	instantaneous rate
$\sigma_{\text{eff}} = \frac{\text{MFLS} \sqrt{M_G}}{2\mu_G \sqrt{E}}$	effective singular value
$\rho = \frac{4\sigma^2 \mu_G^2}{M_G} (1 - \gamma_{\text{max}})$	theoretical rate
$\Psi^* = \frac{\theta \cdot \text{MFLS}}{2\sqrt{E}}$	admissibility threshold
$\mathcal{A} = \frac{\text{MFLS}}{\sqrt{E}} = \frac{\ g_X\ }{\sqrt{E}}$	admissibility ratio
$\text{SafetyRatio} = \frac{\theta \cdot \text{MFLS}}{2M_{\text{max}} \sqrt{E}}$	real-time safety signal
$\cos \theta_t = \frac{\langle F_{\text{base}}, g_X \rangle}{\ F_{\text{base}}\ \ g_X\ }$	alignment
$\cos \psi_t = \frac{\langle \tilde{G}_t, g_t \rangle_F}{\left\ \tilde{G}_t \right\ _F \ g_t\ }$	misalignment (sixth signal)
$\mathcal{R} = \frac{1}{2} A^{-1} (J J^\top)^{-1} J$	channel recovery operator
$S_t = \mathcal{R}(X_t) g_X(t)$	channel recovery formula
$a_k = \tilde{S}_k^2 / \left\ \tilde{S} \right\ ^2$	channel attribution

20.2 The two-object reduction

Everything derives from g_X and E .

g_X carries: MFLS , σ_{eff} , Ψ^* , SafetyRatio , $\cos \theta_t$, channel recovery.

E carries: γ , ρ_{eff} , $\tau_{1/2}$, admissibility condition.

The entire framework is a set of scalar and matrix operations on two vectors computed once per

step.

Part II: Per-Channel Derivations

This part derives every BSDT channel formula from first principles: the channel scalar, its Jacobian row, its contribution to the assembled Jacobian J , its per-channel energy, MFLS, Lyapunov contribution, attribution, and domain instantiations.

21 Calibration Objects

21.1 Normal-period statistics

All channels are measured as deviations from a calibrated normal period. The calibration objects are fitted once on T_0 steps of normal-period data $\{X_t\}_{t=1}^{T_0}$ and then frozen.

Definition 21.1 (Global calibration mean).

$$\mu_0 = \frac{1}{T_0 N} \sum_{t=1}^{T_0} \sum_{i=1}^N x_t^{(i)} \in \mathbb{R}^d$$

Definition 21.2 (Normal-period covariance).

$$\Sigma_0 = \frac{1}{T_0 N} \sum_{t=1}^{T_0} \sum_{i=1}^N (x_t^{(i)} - \mu_0)(x_t^{(i)} - \mu_0)^\top \in \mathbb{R}^{d \times d}$$

Definition 21.3 (Centred state matrix).

$$\tilde{X}_t = X_t - \mathbf{1}_N \mu_0^\top \in \mathbb{R}^{N \times d}$$

Row i : $\tilde{x}_t^{(i)} = x_t^{(i)} - \mu_0$.

Definition 21.4 (PCA basis). Let $V_k \in \mathbb{R}^{d \times k}$ be the matrix of top- k eigenvectors of Σ_0 , i.e. $\Sigma_0 V_k = V_k \Lambda_k$ where $\Lambda_k = \text{diag}(\lambda_1, \dots, \lambda_k)$ with $\lambda_1 \geq \dots \geq \lambda_k$. Default: $k = 4$.

Definition 21.5 (Normal-period velocity threshold).

$$v_0 = \text{Pct}_{95} \left\{ \left\| x_t^{(i)} - x_{t-1}^{(i)} \right\| : t = 1, \dots, T_0, i = 1, \dots, N \right\}$$

The 95th-percentile bar-to-bar velocity during normal period.

22 Channel 1: Camouflage δ_C

22.1 Formula

$$\delta_C^{(i)} = (x_t^{(i)} - \mu_0)^\top \Sigma_0^{-1} (x_t^{(i)} - \mu_0) = \left\| \tilde{x}_t^{(i)} \right\|_{\Sigma_0^{-1}}^2$$

22.2 Derivation

The Mahalanobis distance from agent i 's state to the normal-period centroid μ_0 , measured in the metric Σ_0^{-1} .

Step 1: Scalar form.

$$\delta_C^{(i)} = \tilde{x}_t^{(i)\top} \Sigma_0^{-1} \tilde{x}_t^{(i)} = \sum_{a,b} [\Sigma_0^{-1}]_{ab} \tilde{x}_{t,a}^{(i)} \tilde{x}_{t,b}^{(i)}.$$

Step 2: Matrix form over all agents.

$$\sum_i \delta_C^{(i)} = \text{tr}(\tilde{X}_t \Sigma_0^{-1} \tilde{X}_t^\top) = \left\| \tilde{X}_t \Sigma_0^{-1/2} \right\|_F^2.$$

Step 3: BSDT energy from camouflage only. With $G = \Sigma_0^{-1}$, $S = \text{vec}(\tilde{X}_t)$, $J = I_{Nd}$ (affine map, constant Jacobian):

$$E_C = S^\top G S = \left\| \tilde{X}_t \Sigma_0^{-1/2} \right\|_F^2 = \sum_i \delta_C^{(i)}.$$

22.3 Jacobian $\partial \delta_C^{(i)} / \partial x_t^{(i)}$

$$\frac{\partial \delta_C^{(i)}}{\partial x_t^{(i)}} = 2 \Sigma_0^{-1} (x_t^{(i)} - \mu_0) = 2 \Sigma_0^{-1} \tilde{x}_t^{(i)} \in \mathbb{R}^d$$

Proof. $\delta_C^{(i)} = \tilde{x}^\top \Sigma_0^{-1} \tilde{x}$ with $\tilde{x} = x_t^{(i)} - \mu_0$. By the quadratic form gradient: $\nabla_x [\tilde{x}^\top A \tilde{x}] = 2A\tilde{x}$ for symmetric A . Here $A = \Sigma_0^{-1}$ is symmetric. $\square$

22.4 Full row 1 of J

The camouflage Jacobian row in $\mathbb{R}^{N \times d}$ (one row per agent, d columns):

$$J_C = 2 \tilde{X}_t \Sigma_0^{-1} \in \mathbb{R}^{N \times d}$$

In index form: $[J_C]_{ij} = 2[\Sigma_0^{-1} \tilde{x}_t^{(i)}]_j$. Agents decouple: row i depends only on agent i 's state.

22.5 Geometric interpretation

The iso- δ_C surfaces are ellipsoids in $\mathbb{R}^d$:

$$\mathcal{E}_c = \{x \in \mathbb{R}^d : (x - \mu_0)^\top \Sigma_0^{-1} (x - \mu_0) = c^2\}$$

Semi-axis k has length $c\sqrt{\lambda_k(\Sigma_0)}$ aligned with eigenvector v_k .

Large δ_C : agent has moved far from the calibration centroid in Mahalanobis metric. $\delta_C = 0$: agent is exactly at μ_0 .

22.6 Statistical interpretation

Under the normal-period model $p_0(x) = \mathcal{N}(\mu_0, \Sigma_0)$:

$$\delta_C^{(i)} = -2 \log \frac{p_0(x_t^{(i)})}{p_0(\mu_0)}.$$

δ_C is twice the negative log-likelihood ratio. Under H_0 (normal regime): $\delta_C^{(i)} \sim \chi_d^2$. Total: $\sum_i \delta_C^{(i)} \sim \chi_{Nd}^2$.

22.7 Domain instantiations

Domain	What δ_G detects
ERCOT	Generator capacity-factor vector outside calibration ellipsoid
FDIC	Bank balance-sheet ratios deviating jointly in calibrated metric
Crypto	Joint price+volume+OI+funding outside normal-period covariance
EEG	Electrode activity pattern outside normal-period brain state
Protein	Residue contact configuration outside folded-state covariance

23 Channel 2: Feature Gap δ_G

23.1 Formula

$$\delta_G^{(i)} = \left\| (I - V_k V_k^\top)(x_t^{(i)} - \mu_0) \right\|^2 = \left\| \tilde{x}_t^{(i)} \right\|^2 - \left\| V_k^\top \tilde{x}_t^{(i)} \right\|^2$$

23.2 Derivation

Step 1: PCA projection decomposition. Any vector $\tilde{x} \in \mathbb{R}^d$ decomposes orthogonally:

$$\tilde{x} = \underbrace{V_k V_k^\top \tilde{x}}_{\text{principal subspace}} + \underbrace{(I - V_k V_k^\top) \tilde{x}}_{\text{residual (gap)}}.$$

Pythagorean identity:

$$\|\tilde{x}\|^2 = \|V_k^\top \tilde{x}\|^2 + \|(I - V_k V_k^\top) \tilde{x}\|^2.$$

Step 2: Define the gap. $\delta_G^{(i)}$ is the squared norm of the residual component — the part of the deviation that lies outside the k -dimensional principal subspace of Σ_0 :

$$\delta_G^{(i)} = \left\| (I - V_k V_k^\top) \tilde{x}_t^{(i)} \right\|^2.$$

Step 3: Alternative form.

$$\delta_G^{(i)} = \left\| \tilde{x}_t^{(i)} \right\|^2 - \left\| V_k^\top \tilde{x}_t^{(i)} \right\|^2 = \tilde{x}_t^{(i)\top} (I - V_k V_k^\top) \tilde{x}_t^{(i)}.$$

The matrix $P_\perp = I - V_k V_k^\top$ is the orthogonal projector onto the complement of the principal subspace.

23.3 Angle of deviation

$$\sin^2 \phi_i = \frac{\delta_G^{(i)}}{\left\| \tilde{x}_t^{(i)} \right\|^2} = 1 - \frac{\left\| V_k^\top \tilde{x}_t^{(i)} \right\|^2}{\left\| \tilde{x}_t^{(i)} \right\|^2}$$

$\phi_i = 0$: agent stays within the principal subspace (historically observed directions). $\phi_i = \pi/2$: deviation entirely orthogonal to the principal subspace (unprecedented direction).

23.4 Jacobian $\partial \delta_G^{(i)} / \partial x_t^{(i)}$

$$\frac{\partial \delta_G^{(i)}}{\partial x_t^{(i)}} = 2(I - V_k V_k^\top) \tilde{x}_t^{(i)} \in \mathbb{R}^d$$

Proof. $\delta_G^{(i)} = \tilde{x}^\top P_\perp \tilde{x}$ with $P_\perp = I - V_k V_k^\top$ symmetric and idempotent. $\nabla_x[\tilde{x}^\top P_\perp \tilde{x}] = 2P_\perp \tilde{x}$. $\square$

23.5 Full row 2 of J

$$J_G = 2\tilde{X}_t(I - V_k V_k^\top) \in \mathbb{R}^{N \times d}$$

23.6 Why δ_G is the most powerful channel

From the FDIC result (Fisher- $w = 0.835$, the highest across all domains): δ_G detects motion into PCA residual directions — the low-variance directions that Σ_0 barely covers.

Novel risk always enters through these directions:

- Quantum: rarely-excited decoherence modes
- Banks: novel asset categories (SVB held-to-maturity bonds)
- Supply chain: non-primary supplier routes (semiconductor 2021)
- Cyber: LOLBin patterns (living-off-the-land binaries)
- Crypto: new funding-rate + basis combination not in calibration

The principal subspace captures what the system normally does. δ_G captures what the system has never done before.

23.7 Relation to δ_C

$$\delta_C^{(i)} = \left\| \tilde{x}_t^{(i)} \right\|_{\Sigma_0^{-1}}^2 = \sum_{k=1}^d \frac{[\tilde{x}_t^{(i)} \cdot v_k]^2}{\lambda_k(\Sigma_0)}.$$

This weights the principal directions heavily (large λ_k) and the gap directions lightly (small λ_k).

$\delta_G^{(i)} = \sum_{k' > k} [\tilde{x}_t^{(i)} \cdot v_{k'}]^2$: unweighted sum over residual directions.

So δ_C is blind to motion in low-variance gap directions (small $\lambda_{k'}$ suppresses the weight). δ_G is specifically sensitive to exactly those directions. They are complementary detectors.

24 Channel 3: Activity Anomaly δ_A

24.1 Formula

$$\delta_A^{(i)} = \max\left(0, \left\| x_t^{(i)} - x_{t-1}^{(i)} \right\| - v_0\right)$$

24.2 Derivation

Step 1: Bar-to-bar velocity.

$$v_t^{(i)} = \left\| x_t^{(i)} - x_{t-1}^{(i)} \right\| \in \mathbb{R}_{\geq 0}.$$

This is the Euclidean speed of agent i from step $t - 1$ to step t .

Step 2: Threshold. $v_0 = \text{Pctl}_{95}\{v_t^{(i)}\}$ over the calibration window. During normal operation, 95% of velocities are at or below v_0 .

Step 3: Excess velocity.

$$\delta_A^{(i)} = \max(0, v_t^{(i)} - v_0) = (v_t^{(i)} - v_0)^+.$$

Zero below threshold. Linear above threshold.

Step 4: Total activity score.

$$\delta_A = \|\max(0, \|X_t - X_{t-1}\|_{\text{row}} - v_0)\|_1 = \sum_i \delta_A^{(i)}.$$

In matrix form: compute row-wise norms of $(X_t - X_{t-1})$, subtract v_0 , clamp, sum.

24.3 Jacobian $\partial \delta_A^{(i)} / \partial x_t^{(i)}$

$$\frac{\partial \delta_A^{(i)}}{\partial x_t^{(i)}} = \frac{x_t^{(i)} - x_{t-1}^{(i)}}{\left\| x_t^{(i)} - x_{t-1}^{(i)} \right\|} \cdot \mathbf{1} \left[\left\| x_t^{(i)} - x_{t-1}^{(i)} \right\| > v_0 \right] \in \mathbb{R}^d$$

Proof. Let $v = \left\| x_t^{(i)} - x_{t-1}^{(i)} \right\|$. When $v > v_0$: $\delta_A^{(i)} = v - v_0$, so $\partial \delta_A^{(i)} / \partial x_t^{(i)} = \partial v / \partial x_t^{(i)} = (x_t^{(i)} - x_{t-1}^{(i)}) / v$. When $v \leq v_0$: $\delta_A^{(i)} = 0$, gradient is zero. $\square$

24.4 Full row 3 of J

$$J_A = \frac{X_t - X_{t-1}}{\|X_t - X_{t-1}\|_{\text{row}}} \odot \mathbf{1}[\|X_t - X_{t-1}\|_{\text{row}} > v_0] \in \mathbb{R}^{N \times d}$$

where division and the indicator are row-wise, and $\odot$ broadcasts over columns.

In index form:

$$[J_A]_{ij} = \frac{[x_t^{(i)} - x_{t-1}^{(i)}]_j}{\left\| x_t^{(i)} - x_{t-1}^{(i)} \right\|} \cdot \mathbf{1} \left[\left\| x_t^{(i)} - x_{t-1}^{(i)} \right\| > v_0 \right].$$

24.5 Geometric meaning of J_A

The gradient of $\delta_A^{(i)}$ is the unit velocity vector when above threshold, zero otherwise. It points in the direction of motion — the direction the agent is currently moving. The canonical correction $g_X^{(A)} = 2[AS]_A J_A^\top$ therefore brakes motion in the current movement direction when velocity is anomalous.

24.6 Requirement: two consecutive snapshots

δ_A requires X_{t-1} as well as X_t . All other channels require only the current snapshot. At $t = 1$ (first step): set $\delta_A = 0$ (no velocity estimate available).

24.7 Domain instantiations

Domain	What δ_A detects
ERCOT	Sudden generation ramp: capacity factor changing too fast
FDIC	Bank metric jumping between quarters (balance-sheet shock)
Crypto	Price/OI/liquidation velocity spiking simultaneously
EEG	Electrode voltage changing faster than normal brain activity
Epidemiology	Infection rate changing faster than reproductive dynamics
Social media	Virality metric accelerating above normal spread rate

25 Channel 4: Temporal Novelty δ_T

25.1 Formula

$$\delta_T^{(i)} = \max\left(0, -\log \hat{p}_h(x_t^{(i)})\right)$$

25.2 KDE density estimate

$$\hat{p}_h(x) = \frac{1}{H} \sum_{s=1}^H \frac{1}{(2\pi h^2)^{d/2}} \exp\left(-\frac{\|x - x_{t-s}^{(i)}\|^2}{2h^2}\right)$$

Parameters:

- H : history window length (default: last 50–200 steps)
- h : bandwidth via Scott’s rule: $h = H^{-1/(d+4)}$
- $x_{t-s}^{(i)}$: agent i ’s own state s steps ago

25.3 Step-by-step derivation

Step 1: Build the self-history kernel. At time t , agent i ’s recent history is $\{x_{t-1}^{(i)}, \dots, x_{t-H}^{(i)}\}$. The KDE places a Gaussian kernel of bandwidth h at each historical state. The density at the current state is the average kernel value.

Step 2: Negative log density.

$$-\log \hat{p}_h(x_t^{(i)}) = -\log \left[\frac{1}{H} \sum_{s=1}^H \frac{e^{-\|x_t^{(i)} - x_{t-s}^{(i)}\|^2 / (2h^2)}}{(2\pi h^2)^{d/2}} \right].$$

High value: current state has low density under recent history = temporally novel. Low value: current state is in a dense region of recent history = familiar.

Step 3: Max clamp. When $\hat{p}_h > 1$: $-\log \hat{p}_h < 0$ — hyper-familiar state. Clamped to zero: no novelty signal when state is over-represented.

Step 4: Bandwidth via Scott’s rule.

$$h = H^{-1/(d+4)}.$$

Derivation: Scott’s rule minimises the asymptotic mean integrated squared error of a Gaussian KDE with H samples in d dimensions. As $H \rightarrow \infty$: $h \rightarrow 0$ (finer resolution with more data). As d increases: h decreases slower (curse of dimensionality compensation).

25.4 Kernel weights

Define the normalised kernel weights:

$$w_s^{(i)} = \frac{\exp\left(-\|x_t^{(i)} - x_{t-s}^{(i)}\|^2 / (2h^2)\right)}{\sum_{s'=1}^H \exp\left(-\|x_t^{(i)} - x_{t-s'}^{(i)}\|^2 / (2h^2)\right)}$$

These are softmax weights over the history. $w_s^{(i)} \in (0, 1)$, $\sum_s w_s^{(i)} = 1$. $w_s^{(i)}$ is large when $x_{t-s}^{(i)}$ is close to $x_t^{(i)}$ — recent visits to nearby states get high weight.

25.5 Jacobian $\partial \delta_T^{(i)} / \partial x_t^{(i)}$

$$\frac{\partial \delta_T^{(i)}}{\partial x_t^{(i)}} = \frac{1}{h^2} \sum_{s=1}^H w_s^{(i)} (x_t^{(i)} - x_{t-s}^{(i)}) \cdot \mathbf{1}[\delta_T^{(i)} > 0] \in \mathbb{R}^d$$

Proof. When $\delta_T^{(i)} > 0$ (i.e. $\hat{p}_h < 1$):

$$\begin{aligned} \delta_T^{(i)} &= -\log \hat{p}_h(x_t^{(i)}). \\ \frac{\partial \delta_T^{(i)}}{\partial x_t^{(i)}} &= -\frac{1}{\hat{p}_h} \frac{\partial \hat{p}_h}{\partial x_t^{(i)}}. \end{aligned}$$

Compute the KDE gradient:

$$\frac{\partial \hat{p}_h}{\partial x_t^{(i)}} = \frac{1}{H} \sum_{s=1}^H \frac{1}{(2\pi h^2)^{d/2}} e^{-\|x_t^{(i)} - x_{t-s}^{(i)}\|^2 / (2h^2)} \cdot \frac{-(x_t^{(i)} - x_{t-s}^{(i)})}{h^2}.$$

Factor out $\hat{p}_h$:

$$\frac{\partial \hat{p}_h}{\partial x_t^{(i)}} = -\frac{\hat{p}_h}{h^2} \sum_{s=1}^H w_s^{(i)} (x_t^{(i)} - x_{t-s}^{(i)}).$$

Therefore:

$$\frac{\partial \delta_T^{(i)}}{\partial x_t^{(i)}} = -\frac{1}{\hat{p}_h} \cdot \left(-\frac{\hat{p}_h}{h^2}\right) \sum_s w_s^{(i)} (x_t^{(i)} - x_{t-s}^{(i)}) = \frac{1}{h^2} \sum_s w_s^{(i)} (x_t^{(i)} - x_{t-s}^{(i)}). \quad \square$$

□

Geometric meaning. The gradient is the weighted average displacement from current state to all recent history states, pointing *away* from the centroid of recent history. It points in the direction of maximum temporal novelty: away from where the agent has recently been.

25.6 Full row 4 of J

$$J_T = \frac{1}{h^2} \sum_{s=1}^H W_s \odot (X_t - X_{t-s}) \cdot \mathbf{1}[\delta_T > 0] \in \mathbb{R}^{N \times d}$$

where $W_s \in \mathbb{R}^{N \times 1}$ contains the per-agent kernel weights $w_s^{(i)}$, broadcast over the d columns via $\odot$.

In index form:

$$[J_T]_{ij} = \frac{1}{h^2} \sum_{s=1}^H w_s^{(i)} \left[x_t^{(i)} - x_{t-s}^{(i)} \right]_j \cdot \mathbf{1}[\delta_T^{(i)} > 0].$$

25.7 Interpretation: δ_T vs δ_C

δ_C measures: how far from the *calibration mean* μ_0 .

δ_T measures: how far from *anything recently visited*.

A state can be near μ_0 (low δ_C) but temporally novel (high δ_T) if the agent has been away from that region for longer than the KDE window. This is the *return-to-calm-but-unfamiliar* regime — dangerous because δ_C gives no alarm while δ_T correctly flags that the dynamics are in uncharted territory.

25.8 Domain instantiations

Domain	What δ_T detects
Supply chain	Lead-time at historically unprecedented combination (+98 wk lead)
Quantum	Fidelity drift at unseen coherence level (dominant channel, Fisher- $w=0.584$)
ERCOT	Grid at frequency+generation mix not seen in KDE history
Crypto	Price+funding+OI at jointly novel state (new regime entry)
Social media	Virality at speed+content combination outside recent history
Climate	AMOC at temperature+CO2+albedo combination beyond prior envelope

26 The Assembled Jacobian J

26.1 Full matrix

$$J(X_t) = \begin{pmatrix} J_C \\ J_G \\ J_A \\ J_T \end{pmatrix} \in \mathbb{R}^{4 \times Nd}$$

where each row block is $\mathbb{R}^{N \times d}$ reshaped to $\mathbb{R}^{1 \times Nd}$.

$$\begin{aligned}
J_C &= 2\tilde{X}_t \Sigma_0^{-1} \\
J_G &= 2\tilde{X}_t (I - V_k V_k^\top) \\
J_A &= \frac{X_t - X_{t-1}}{\|X_t - X_{t-1}\|_{\text{row}}} \odot \mathbf{1}[v > v_0] \\
J_T &= \frac{1}{h^2} \sum_{s=1}^H W_s \odot (X_t - X_{t-s}) \cdot \mathbf{1}[\delta_T > 0]
\end{aligned}$$

26.2 Gram matrix

$$\mathcal{G} = J J^\top = \begin{pmatrix} \langle J_C, J_C \rangle_F & \langle J_C, J_G \rangle_F & \langle J_C, J_A \rangle_F & \langle J_C, J_T \rangle_F \\ \langle J_G, J_C \rangle_F & \langle J_G, J_G \rangle_F & \langle J_G, J_A \rangle_F & \langle J_G, J_T \rangle_F \\ \langle J_A, J_C \rangle_F & \langle J_A, J_G \rangle_F & \langle J_A, J_A \rangle_F & \langle J_A, J_T \rangle_F \\ \langle J_T, J_C \rangle_F & \langle J_T, J_G \rangle_F & \langle J_T, J_A \rangle_F & \langle J_T, J_T \rangle_F \end{pmatrix} \in \mathbb{R}^{4 \times 4}$$

where $\langle J_a, J_b \rangle_F = \text{tr}(J_a J_b^\top)$.

Diagonal entries (self-inner-products):

$$\begin{aligned}
[J J^\top]_{CC} &= 4 \text{tr}(\tilde{X}_t \Sigma_0^{-1} \Sigma_0^{-1} \tilde{X}_t^\top) = 4 \left\| \tilde{X}_t \Sigma_0^{-1} \right\|_F^2 \\
[J J^\top]_{GG} &= 4 \text{tr}(\tilde{X}_t (I - V_k V_k^\top)^2 \tilde{X}_t^\top) = 4 \left\| \tilde{X}_t (I - V_k V_k^\top) \right\|_F^2 \\
[J J^\top]_{AA} &= \text{tr} \left(\frac{(X_t - X_{t-1})(X_t - X_{t-1})^\top}{\|X_t - X_{t-1}\|_{\text{row}}^2} \cdot \mathbf{1}[v > v_0] \right) \\
[J J^\top]_{TT} &= \frac{1}{h^4} \left\| \sum_s W_s \odot (X_t - X_{t-s}) \cdot \mathbf{1}[\delta_T > 0] \right\|_F^2
\end{aligned}$$

Cross terms (C-G):

$$[J J^\top]_{CG} = 4 \text{tr}(\tilde{X}_t \Sigma_0^{-1} (I - V_k V_k^\top) \tilde{X}_t^\top).$$

Note: $(I - V_k V_k^\top)$ projects onto the PCA gap; Σ_0^{-1} weights by inverse variance. These are not orthogonal in general — channels δ_C and δ_G share information unless Σ_0 has uniform spectrum.

26.3 When channels are orthogonal

$[J J^\top]_{CG} = 0$ iff $\text{tr}(\tilde{X}_t \Sigma_0^{-1} (I - V_k V_k^\top) \tilde{X}_t^\top) = 0$, which holds when the state deviation $\tilde{X}_t$ has no overlap between the Σ_0^{-1} -weighted direction and the gap-projected direction.

In the case $k = d$ (all PCA components retained): $I - V_k V_k^\top = 0$, so $J_G = 0$ and $\delta_G = 0$ — the gap channel vanishes when all directions are principal.

In the case $k = 0$: $I - V_k V_k^\top = I$, so $J_G = 2\tilde{X}_t$ — the gap channel measures total Euclidean deviation, unweighted.

The identifiability condition (Theorem R.1): $J J^\top \succ 0$ requires all four channels to be linearly independent in $\mathbb{R}^{Nd}$. This is generically satisfied when: $k < d$ (PCA is a strict subspace), $v_t^{(i)} > v_0$ for at least one agent (activity active), and $\delta_T^{(i)} > 0$ for at least one agent (novelty active).

27 Per-Channel Canonical Gradient

27.1 Decomposition

The canonical gradient decomposes exactly over channels:

$$g_X = 2J^\top GS = 2 \sum_{k \in \{C, G, A, T\}} [AS]_k J_k^\top$$

where $[AS]_k$ is the k -th component of $AS \in \mathbb{R}^4$.

$$\textit{Proof. } g_X = 2J^\top GS = 2 \begin{pmatrix} J_C^\top & J_G^\top & J_A^\top & J_T^\top \end{pmatrix} \begin{pmatrix} [GS]_C \\ [GS]_G \\ [GS]_A \\ [GS]_T \end{pmatrix} = 2 \sum_k [GS]_k J_k^\top. \quad \square$$

27.2 Per-channel gradient vectors

$$\begin{aligned} g_X^{(C)} &= 2[AS]_C \cdot J_C^\top = 4[AS]_C \Sigma_0^{-1} \tilde{X}_t^\top \in \mathbb{R}^{Nd} \\ g_X^{(G)} &= 2[AS]_G \cdot J_G^\top = 4[AS]_G (I - V_k V_k^\top) \tilde{X}_t^\top \in \mathbb{R}^{Nd} \\ g_X^{(A)} &= 2[AS]_A \cdot J_A^\top \in \mathbb{R}^{Nd} \\ g_X^{(T)} &= 2[AS]_T \cdot J_T^\top \in \mathbb{R}^{Nd} \end{aligned}$$

27.3 MFLS per channel

$$\text{MFLS}^{(k)} = \|g_X^{(k)}\| = 2|[AS]_k| \cdot \|J_k\|_F$$

$$\textit{Proof. } \|g_X^{(k)}\| = \|2[AS]_k J_k^\top\| = 2|[AS]_k| \|J_k\|_F. \quad \square$$

Expanded per channel:

$$\begin{aligned} \text{MFLS}^{(C)} &= 4|[AS]_C| \cdot \|\tilde{X}_t \Sigma_0^{-1}\|_F \\ \text{MFLS}^{(G)} &= 4|[AS]_G| \cdot \|\tilde{X}_t (I - V_k V_k^\top)\|_F \\ \text{MFLS}^{(A)} &= 2|[AS]_A| \cdot \|J_A\|_F \\ \text{MFLS}^{(T)} &= 2|[AS]_T| \cdot \|J_T\|_F \end{aligned}$$

Note: $\text{MFLS} \neq \sum_k \text{MFLS}^{(k)}$ in general (triangle inequality); the correct relationship is:

$$\text{MFLS}^2 = \left\| \sum_k g_X^{(k)} \right\|^2 = \sum_{k, k'} \langle g_X^{(k)}, g_X^{(k')} \rangle.$$

28 Per-Channel Energy Contributions

28.1 Energy decomposition

The total energy under coupling matrix A :

$$E = S^\top A S = \sum_{k,k'} A_{kk'} S_k S_{k'}.$$

The diagonal contribution of channel k :

$$E^{(k)} = A_{kk} S_k^2.$$

The cross-channel coupling:

$$E^{(k,k')} = 2A_{kk'} S_k S_{k'}, \quad k \neq k'.$$

Total:

$$E = \sum_k E^{(k)} + \sum_{k < k'} E^{(k,k')}.$$

When $A = I$ (identity coupling): channels decouple, $E = \sum_k S_k^2 = \delta_C^2 + \delta_G^2 + \delta_A^2 + \delta_T^2$.

When $A = \text{diag}(a_C, a_G, a_A, a_T)$: channels still decouple, $E = a_C \delta_C^2 + a_G \delta_G^2 + a_A \delta_A^2 + a_T \delta_T^2$.

29 Per-Channel Lyapunov Contributions $\dot{V}_k$

29.1 Decomposition theorem

Theorem 29.1 (Per-channel Lyapunov, §XXV of anatomy document).

$$\dot{V} = \frac{dE}{dt} = \sum_{k \in \{C, G, A, T\}} \dot{V}_k$$

where

$$\dot{V}_k = [g_t]_k \left[\langle \partial_X \delta_k, F_t \rangle_F - \gamma_t^* \frac{\langle F_t, g_t \rangle}{\|g_t\|^2} \langle \partial_X \delta_k, g_t \rangle_F \right] = \dot{V}_k^{\text{drift}} + \dot{V}_k^{\text{control}}$$

Proof. $\dot{V} = \langle \tilde{G}_t, \dot{X} \rangle_F$ where $\tilde{G}_t = J^\top g_t = \sum_k [g_t]_k J_k^\top$. Substituting the canonical ODE $\dot{X} = F_t - \gamma_t^* \alpha_t g_t$:

$$\dot{V} = \langle \tilde{G}_t, F_t \rangle_F - \gamma_t^* \alpha_t \langle \tilde{G}_t, g_t \rangle_F.$$

Expanding $\tilde{G}_t = \sum_k [g_t]_k J_k^\top$:

$$\dot{V} = \sum_k [g_t]_k \langle J_k^\top, F_t \rangle_F - \gamma_t^* \alpha_t \sum_k [g_t]_k \langle J_k^\top, g_t \rangle_F = \sum_k \dot{V}_k. \quad \square$$

□

29.2 Drift and control components

$$\begin{aligned} \dot{V}_k^{\text{drift}} &= [g_t]_k \langle J_k^\top, F_t \rangle_F = [g_t]_k \langle \partial_X \delta_k, F_t \rangle_F \\ \dot{V}_k^{\text{control}} &= -[g_t]_k \gamma_t^* \frac{\langle F_t, g_t \rangle}{\|g_t\|^2} \langle J_k^\top, g_t \rangle_F \end{aligned}$$

29.3 Channel stability attribution

$$a_k^{\text{stab}} = \frac{\max(0, \dot{V}_k)}{\sum_{k'} \max(0, \dot{V}_{k'})}, \quad \sum_k a_k^{\text{stab}} = 1$$

a_k^{stab} is the fraction of total instability rate driven by channel k . This is different from the energy attribution $a_k = \tilde{S}_k^2 / \|\tilde{S}\|^2$: a channel can have high deviation score but contribute little to instability if its spatial gradient is perpendicular to the force field.

29.4 Control efficiency per channel

$$\eta_k = \frac{-\dot{V}_k^{\text{control}}}{\dot{V}_k^{\text{drift}}} = \gamma_t^* \frac{\langle F_t, g_t \rangle \langle J_k^\top, g_t \rangle_F}{\|g_t\|^2 \langle J_k^\top, F_t \rangle_F}$$

$\eta_k \in (0, 1)$: control partially offsets channel k 's drift.

$\eta_k > 1$: control overcorrects channel k .

$\eta_k < 0$: the control is making channel k worse — the projection direction is misaligned with channel k 's spatial gradient. This is the per-channel manifestation of the uncontrollable regime.

30 Per-Channel Attribution a_k

30.1 Energy attribution

$$a_k(t) = \frac{\tilde{S}_k(t)^2}{\|\tilde{S}(t)\|^2}, \quad \tilde{S}_k = \frac{S_k}{\mu_k^{\text{normal}}}, \quad \sum_k a_k = 1$$

The mean-normalisation $\mu_k^{\text{normal}} = \mathbb{E}_{\text{calm}}[S_k]$ is computed once on the calibration window and frozen.

Derivation. Raw channel scores S_k have different scales (δ_C is chi-squared, δ_T is log-density, etc.). Mean-normalisation converts each to a dimensionless multiple of its normal-period average. The squared normalised score is then the fraction of total anomaly energy carried by channel k .

30.2 MFLS attribution

The fraction of total MFLS squared driven by channel k :

$$c_k = \frac{[\nabla_S E_{\text{BS}}]_k^2}{\|\nabla_S E_{\text{BS}}\|^2} = \frac{[g_t]_k^2}{\|g_t\|^2}.$$

This is the channel-space attribution — different from a_k (energy attribution) and a_k^{stab} (stability attribution).

The three attributions agree only when channels are orthogonal and the coupling matrix A is diagonal.

30.3 Dominant channel

$$k^* = \arg \max_k a_k(t)$$

identifies the primary driver of current risk.

Empirical dominant channels across domains:

Domain	Dominant k^*	Fisher- w
ERCOT	δ_C (camouflage)	0.623
FDIC	δ_G (feature gap)	0.835
World Bank	δ_G (capital gap)	0.320
Supply chain	δ_T (lead-time novelty)	0.433
Quantum	δ_T (fidelity drift)	0.584
EEG	δ_A (activity)	0.327
Climate	δ_A (activity)	0.407
Epidemiology	δ_A (activity)	0.518
Social media	δ_A (activity)	0.510

Pattern: δ_A dominates cascade-saturation systems; δ_G dominates structural-exhaustion systems; δ_T dominates regime-shift systems.

31 Control Coupling Sum R_t

$$R_t = \sum_{k \in \{C, G, A, T\}} [g_t]_k \langle J_k^\top, g_t \rangle_F = \langle \tilde{G}_t, g_t \rangle_F = \left\| \tilde{G}_t \right\|_F \|g_t\| \cos \psi_t$$

Derivation. $R_t = \langle J^\top g_t, g_t \rangle_F = g_t^\top J J^\top g_t = g_t^\top \mathcal{G} g_t \geq 0$ when $\mathcal{G} \succ 0$.

$R_t > 0$: control direction positively coupled to all channel spatial gradients. $R_t \leq 0$: uncontrollable regime — control direction orthogonal to or opposed by channel gradients.

Per-channel expansion:

$$\begin{aligned} R_t^{(C)} &= [g_t]_C \langle J_C^\top, g_t \rangle_F = 4[g_t]_C \langle \tilde{X}_t \Sigma_0^{-1}, g_t \rangle_F \\ R_t^{(G)} &= [g_t]_G \langle J_G^\top, g_t \rangle_F = 4[g_t]_G \langle \tilde{X}_t (I - V_k V_k^\top), g_t \rangle_F \\ R_t^{(A)} &= [g_t]_A \langle J_A^\top, g_t \rangle_F \\ R_t^{(T)} &= [g_t]_T \langle J_T^\top, g_t \rangle_F \end{aligned}$$

$$R_t = \sum_k R_t^{(k)}.$$

32 Complete Per-Step Algorithm

32.1 All objects from raw data

Given $X_t, X_{t-1}, \{X_{t-s}\}_{s=1}^H$ and calibration $(\mu_0, \Sigma_0^{-1}, \Sigma_0^{-1/2}, V_k, v_0, A, \theta)$:

$$1. \tilde{X}_t = X_t - \mathbf{1} \mu_0^\top$$

2. **Channel scalars:**

$$\begin{aligned}\delta_C^{(i)} &= \tilde{x}_t^{(i)\top} \Sigma_0^{-1} \tilde{x}_t^{(i)} \quad \forall i \\ \delta_G^{(i)} &= \left\| (I - V_k V_k^\top) \tilde{x}_t^{(i)} \right\|^2 \quad \forall i \\ \delta_A^{(i)} &= \max(0, \left\| x_t^{(i)} - x_{t-1}^{(i)} \right\| - v_0) \quad \forall i \\ \delta_T^{(i)} &= \max(0, -\log \hat{p}_h(x_t^{(i)})) \quad \forall i\end{aligned}$$

3. $S_t = (\bar{\delta}_C, \bar{\delta}_G, \bar{\delta}_A, \bar{\delta}_T)^\top \in \mathbb{R}^4$ where $\bar{\delta}_k = \sum_i \delta_k^{(i)} / N$ (mean over agents)

4. **Jacobian rows:**

$$\begin{aligned}J_C &= 2\tilde{X}_t \Sigma_0^{-1} \\ J_G &= 2\tilde{X}_t (I - V_k V_k^\top) \\ J_A &= \frac{X_t - X_{t-1}}{\|X_t - X_{t-1}\|_{\text{row}}} \odot \mathbf{1}[v > v_0] \\ J_T &= \frac{1}{h^2} \sum_s W_s \odot (X_t - X_{t-s}) \cdot \mathbf{1}[\delta_T > 0]\end{aligned}$$

5. $J = [J_C; J_G; J_A; J_T] \in \mathbb{R}^{4 \times Nd}$

6. $\mathcal{G} = J J^\top \in \mathbb{R}^{4 \times 4}$

7. $E_t = S_t^\top A S_t$

8. $g_X = 2J^\top A S_t \in \mathbb{R}^{Nd}$ (canonical gradient = MFLS direction)

9. $\text{MFLS}_t = \|g_X\|$

10. $\gamma_t = E_t / (E_t + \theta)$

11. $\rho_{\text{eff}}(t) = \text{MFLS}_t^2 (1 - \gamma_t) / E_t$

12. $\Psi^*(t) = \theta \cdot \text{MFLS}_t / (2\sqrt{E_t})$

13. $\text{SafetyRatio}(t) = \Psi^*(t) / M_{\max}$

14. $F_t = -\alpha \tilde{X}_t - L_t X_t$ (GravityEngine force)

15. $\cos \theta_t = \langle F_t, g_X \rangle / (\|F_t\| \text{MFLS}_t)$

16. $\mathcal{R}_t = \frac{1}{2} A^{-1} \mathcal{G}^{-1} J$ (4×4 inversion)

17. $\hat{S}_t = \mathcal{R}_t g_X$ (channel recovery check)

18. $a_k(t) = \tilde{S}_k(t)^2 / \left\| \tilde{S}_t \right\|^2$ (energy attribution)

19. $g_t = 2A S_t + (\beta_k e^{\beta_k S_k})_k + w$ (full channel gradient)

20. $\tilde{G}_t = J^\top g_t$ (pullback gradient)

21. $R_t = \langle \tilde{G}_t, g_t \rangle_F = g_t^\top \mathcal{G} g_t$ (control coupling)

22. $\cos \psi_t = R_t / (\left\| \tilde{G}_t \right\|_F \|g_t\|)$ (misalignment angle)

23. $\dot{V}_k = [g_t]_k [\langle J_k^\top, F_t \rangle_F - \gamma_t^* \alpha_t \langle J_k^\top, g_t \rangle_F]$ (per-channel Lyapunov)
24. $a_k^{\text{stab}}(t) = \max(0, \dot{V}_k) / \sum_{k'} \max(0, \dot{V}_{k'})$ (stability attribution)
25. $\eta_k = -\dot{V}_k^{\text{control}} / \dot{V}_k^{\text{drift}}$ (control efficiency)
26. ODE step: $\dot{X} = F_t - \gamma_t \alpha_t g_X$
27. $X_{t+1} = X_t + h \dot{X}$

33 Complete Formula Reference

Channels

$$\begin{aligned}\delta_C^{(i)} &= (x_t^{(i)} - \mu_0)^\top \Sigma_0^{-1} (x_t^{(i)} - \mu_0) \\ \delta_G^{(i)} &= \left\| (I - V_k V_k^\top) (x_t^{(i)} - \mu_0) \right\|^2 \\ \delta_A^{(i)} &= \max(0, \left\| x_t^{(i)} - x_{t-1}^{(i)} \right\| - v_0) \\ \delta_T^{(i)} &= \max(0, -\log \hat{p}_h(x_t^{(i)}))\end{aligned}$$

Jacobian rows

$$\begin{aligned}J_C &= 2\tilde{X}_t \Sigma_0^{-1} \\ J_G &= 2\tilde{X}_t (I - V_k V_k^\top) \\ J_A &= \frac{X_t - X_{t-1}}{\|X_t - X_{t-1}\|_{\text{row}}} \odot \mathbf{1}[v > v_0] \\ J_T &= \frac{1}{h^2} \sum_{s=1}^H W_s \odot (X_t - X_{t-s}) \cdot \mathbf{1}[\delta_T > 0]\end{aligned}$$

Assembled objects

$$\begin{aligned}J &= [J_C; J_G; J_A; J_T] \in \mathbb{R}^{4 \times Nd} \\ \mathcal{G} &= J J^\top \in \mathbb{R}^{4 \times 4} \\ g_X &= 2J^\top G S \quad (= \text{MFLS direction}) \\ \text{MFLS} &= \|g_X\| \\ \mathcal{R} &= \frac{1}{2} A^{-1} \mathcal{G}^{-1} J \quad (4 \times 4 \text{ inversion only})\end{aligned}$$

Per-channel

$$\begin{aligned}
g_X^{(k)} &= 2[AS]_k J_k^\top \\
\text{MFLS}^{(k)} &= 2|[AS]_k| \cdot \|J_k\|_F \\
\dot{V}_k &= [g_t]_k \langle J_k^\top, F_t \rangle_F - \gamma^* \alpha [g_t]_k \langle J_k^\top, g_t \rangle_F \\
a_k &= \tilde{S}_k^2 / \|\tilde{S}\|^2 \quad (\text{energy}) \\
a_k^{\text{stab}} &= \max(0, \dot{V}_k) / \sum_{k'} \max(0, \dot{V}_{k'}) \quad (\text{stability}) \\
\eta_k &= -\dot{V}_k^{\text{control}} / \dot{V}_k^{\text{drift}} \quad (\text{efficiency}) \\
R_t^{(k)} &= [g_t]_k \langle J_k^\top, g_t \rangle_F \quad (\text{coupling})
\end{aligned}$$

Aggregate

$$\begin{aligned}
R_t &= \sum_k R_t^{(k)} = g_t^\top \mathcal{G} g_t \\
\cos \psi_t &= R_t / (\|\tilde{G}_t\|_F \|g_t\|) \\
\Psi^* &= \theta \cdot \text{MFLS} / (2\sqrt{E}) \\
\text{SafetyRatio} &= \theta \cdot \text{MFLS} / (2M_{\max} \sqrt{E})
\end{aligned}$$

Part III: Game-Theoretic Architecture

This part derives the game-theoretic foundations of the canonical engine. Every claim is stated as a theorem with explicit hypotheses, explicit equilibrium concept, and explicit scope. Heuristic interpretations are clearly labelled as such.

34 The Projection Game

34.1 Setup

Fix state X with $g_X \neq 0$. Two players in a **zero-sum game**:

- **Controller** chooses update direction $u \in \mathbb{R}^n$, **minimises** payoff
- **Auditor** chooses penalty multiplier $\lambda \in \mathbb{R}$, **maximises** payoff

The Lagrangian:

$$\mathcal{L}(u, \lambda) = \frac{1}{2} \|u - F_{\text{base}}\|^2 + \lambda \langle u, g_X \rangle - \frac{\theta \|g_X\|^2}{2E} \lambda^2$$

Sign convention. Controller minimises (deviation from F_{base} is costly). Auditor maximises (greater penalty extracts more from energy-injecting updates, at quadratic cost). The game is zero-sum: Controller's loss = Auditor's gain = $\mathcal{L}$.

34.2 Saddle-point theorem

Theorem 34.1 (Unique saddle point, GT.1). *The Lagrangian $\mathcal{L}(u, \lambda)$ has a unique saddle point (u^*, λ^*) with:*

$$u^* = F_{\text{base}} - \gamma \frac{\langle F_{\text{base}}, g_X \rangle}{\|g_X\|^2} g_X, \quad \lambda^* = \frac{\langle F_{\text{base}}, g_X \rangle E}{(E + \theta) \|g_X\|^2}, \quad \gamma = \frac{E}{E + \theta}.$$

Existence and uniqueness. $\partial_u^2 \mathcal{L} = I_n \succ 0$: strictly convex in u . $\partial_\lambda^2 \mathcal{L} = -\theta \|g_X\|^2 / E < 0$: strictly concave in λ . By the Sion minimax theorem (Sion, 1958), a unique saddle point exists for convex-concave problems. $\square$

Formula. Controller's first-order condition:

$$\partial_u \mathcal{L} = (u - F_{\text{base}}) + \lambda g_X = 0 \implies u^* = F_{\text{base}} - \lambda g_X.$$

Auditor's first-order condition:

$$\partial_\lambda \mathcal{L} = \langle u, g_X \rangle - \theta \|g_X\|^2 \lambda / E = 0 \implies \lambda^* = \frac{\langle u^*, g_X \rangle E}{\theta \|g_X\|^2}.$$

Substituting u^* :

$$\lambda^* \theta \|g_X\|^2 / E = \langle F_{\text{base}}, g_X \rangle - \lambda^* \|g_X\|^2 \implies \lambda^* = \frac{\langle F_{\text{base}}, g_X \rangle E}{(E + \theta) \|g_X\|^2}.$$

Substituting back gives $u^* = F_{\text{base}} - \gamma \langle F_{\text{base}}, g_X \rangle g_X / \|g_X\|^2$ with $\gamma = E / (E + \theta)$. $\square$

34.3 Interpretation

The saddle-point game derives the canonical step from a variational principle:

- The Fermi-Dirac gain $\gamma = E/(E + \theta)$ is *not* a chosen damping rule. It is the equilibrium penalty intensity in the projection game.
- θ is the Auditor's quadratic intervention cost. Small θ : aggressive intervention. Large θ : mild intervention.
- The locked canonical ODE step is the unique Nash equilibrium of a well-posed convex-concave zero-sum game.

35 The Adversarial Disturbance Game

35.1 Setup

$F_{\text{base}} = F_{\text{nom}} + F_{\text{adv}}$ where adversary chooses F_{adv} with $\|F_{\text{adv}}\| \leq M_{\text{adv}}$.

- **Controller** chooses $\gamma \in [0, 1)$, minimises $\dot{E}$
- **Adversary** chooses F_{adv} with $\|F_{\text{adv}}\| \leq M_{\text{adv}}$, maximises $\dot{E}$

Payoff (to Adversary):

$$J(\gamma, F_{\text{adv}}) = \dot{E} = (1 - \gamma) \left[\langle g_X, F_{\text{nom}} + F_{\text{adv}} \rangle - \|g_X\|^2 \right].$$

35.2 No interior saddle

Theorem 35.1 (Minimax value, no interior saddle, GT.2). *The minimax value is:*

$$V^* = \inf_{\gamma \in [0, 1)} \sup_{\|F_{\text{adv}}\| \leq M_{\text{adv}}} J(\gamma, F_{\text{adv}}) = \left[\langle g_X, F_{\text{nom}} \rangle + M_{\text{adv}} \|g_X\| - \|g_X\|^2 \right]^+$$

with the adversary's unique maximiser:

$$F_{\text{adv}}^* = M_{\text{adv}} \frac{g_X}{\|g_X\|}.$$

The Controller's minimiser is boundary:

- If $B \leq 0$: $\gamma^* = 0$ attains the value
- If $B > 0$: infimum $V^* = 0$ approached as $\gamma \rightarrow 1^-$, not attained in $[0, 1)$

where $B = \langle g_X, F_{\text{nom}} \rangle + M_{\text{adv}} \|g_X\| - \|g_X\|^2$.

Proof. Adversary's maximisation by Cauchy-Schwarz: $\sup_{F_{\text{adv}}} \langle g_X, F_{\text{adv}} \rangle = M_{\text{adv}} \|g_X\|$ at F_{adv}^* unique on the sphere.

Inner maximum: $\sup_{F_{\text{adv}}} J = (1 - \gamma)B$.

Outer minimisation: linear in $(1 - \gamma)$. If $B \leq 0$: minimum at $\gamma = 0$. If $B > 0$: infimum at $\gamma \rightarrow 1^-$, not attained in the open interval $[0, 1)$. $\square$

Remark 35.2 (Operational degeneracy). The minimax value collapses to 0 via $\gamma \rightarrow 1^-$. This corresponds to the Controller freezing dynamics, which is not operationally meaningful. The unconstrained adversarial game is therefore degenerate as a control-design principle. This motivates the regularised formulation of §35.

36 The Regularised Adversarial Game

36.1 Motivating the regulariser

Pure minimax has no interior solution. Add a regulariser $-\mu\gamma^2$ penalising large gain values (operationally extreme intervention). The coefficient $\mu = \theta \|g_X\|^2 / (2E)$ is chosen so that the equilibrium reduces to the canonical Fermi-Dirac form when $M_{\text{adv}} = 0$, recovering Theorem GT.1 as a special case.

36.2 Regularised game

$$\mathcal{L}_{\text{reg}}(\gamma, F_{\text{adv}}) = (1 - \gamma) \left[\langle g_X, F_{\text{nom}} + F_{\text{adv}} \rangle - \|g_X\|^2 \right] - \mu\gamma^2$$

Theorem 36.1 (Interior equilibrium, GT.3). *With $\mu = \theta \|g_X\|^2 / (2E)$, the regularised game has a unique interior Nash equilibrium at:*

$$\gamma_{\text{rob}} = \frac{E + M_{\text{adv}} \|g_X\|}{E + M_{\text{adv}} \|g_X\| + \theta}, \quad F_{\text{adv}}^* = M_{\text{adv}} \frac{g_X}{\|g_X\|}.$$

Proof. Adversary's maximiser: $F_{\text{adv}}^* = M_{\text{adv}} g_X / \|g_X\|$ by Cauchy–Schwarz.

Define effective energy $\tilde{E} = E + M_{\text{adv}} \|g_X\|$. Controller's first-order condition in γ (with the disturbance absorbed):

$$\partial_\gamma \mathcal{L}_{\text{reg}} = 0 \implies \gamma_{\text{rob}} = \frac{\tilde{E}}{\tilde{E} + \theta} = \frac{E + M_{\text{adv}} \|g_X\|}{E + M_{\text{adv}} \|g_X\| + \theta}.$$

Verification: $M_{\text{adv}} = 0 \implies \tilde{E} = E \implies \gamma_{\text{rob}} = \gamma_{\text{canonical}}$. $\square$

Remark 36.2. The regulariser converts the degenerate minimax problem into a well-posed mechanism-design problem with an interior solution. The robust gain γ_{rob} is the equilibrium of this regularised game.

37 The Multi-Engine Cooperative Game

37.1 Setup

m canonical engines with energies $E_1, \dots, E_m$, gains $\gamma_i = E_i / (E_i + \theta_i)$, gradients $g_X^{(i)}$, damping matrices $R_i = \gamma_i g_X^{(i)} g_X^{(i)\top} / \|g_X^{(i)}\|^2 \succeq 0$.

Composite Hamiltonian: $H(X) = \frac{1}{2} \sum_i E_i$.

Composite dynamics (Definition P.1 of v4):

$$\dot{X} = \sum_i F_{\text{base}}^{(i)} - 2 \sum_i R_i \nabla H.$$

37.2 Required assumption

Assumption GT.4 (Non-positive cross-coupling). For all $i \neq j$:

$$\langle g_X^{(i)}, g_X^{(j)} \rangle \leq 0 \quad \text{or} \quad g_X^{(i)} \perp g_X^{(j)}.$$

Equivalently: the gradient interaction matrix $K_{ij} = \langle g_X^{(i)}, g_X^{(j)} \rangle$ has non-positive off-diagonal entries, or is diagonal (orthogonal engines).

37.3 Conditional superadditivity

Theorem 37.1 (Coalition superadditivity, conditional, GT.4). Under Assumption GT.4, for any partition Π of $\{1, \dots, m\}$:

$$v(\{1, \dots, m\}) \geq \sum_{S \in \Pi} v(S).$$

Proof. $\sum_i \dot{E}_i = -2\langle \nabla H, \sum_j R_j \nabla H \rangle + \text{drift terms}.$

Each $\langle \nabla H, R_j \nabla H \rangle = \gamma_j \langle \nabla H, g_X^{(j)} \rangle^2 / \|g_X^{(j)}\|^2 \geq 0.$

Under Assumption GT.4, cross-coupling terms $\langle g_X^{(i)}, g_X^{(j)} \rangle \leq 0$ contribute negatively to $\sum_i \dot{E}_i$, adding to the dissipation. Hence:

$$\sum_i \dot{E}_i \leq \sum_i \dot{E}_i|_{\text{isolated}} - (\text{non-negative cross terms}).$$

The grand coalition strictly improves (or matches) the partitioned outcome. $\square$

Remark 37.2 (When Assumption GT.4 holds). Assumption GT.4 holds in three important cases:

1. Orthogonal engines: distinct asset classes, distinct sensor channels, distinct control axes
2. Mean-reverting engines: all $F_{\text{base}}^{(i)} = -\kappa_i(X - X_i^*)$ with non-conflicting equilibria
3. Hierarchical engines: outer engine's gradient is in the kernel of inner engine's gradient

It can fail when engines have directly competing objectives. The theorem is conditional, not universal.

38 Channel Attribution as Shapley Value

38.1 Setup

The canonical gradient decomposes by channel:

$$g_X = \sum_{k \in \{C, G, A, T\}} g_X^{(k)}, \quad g_X^{(k)} = 2[AS]_k J_k^\top.$$

Each channel is a player contributing to the coalitional value (gradient magnitude).

38.2 Shapley axioms

A channel attribution $\phi : \{C, G, A, T\} \rightarrow \mathbb{R}$ satisfies:

- (SH1) **Efficiency:** $\sum_k \phi_k = \|g_X\|^2$
- (SH2) **Symmetry:** channels with identical marginal contributions get identical attribution
- (SH3) **Null:** if $g_X^{(k)} = 0$, then $\phi_k = 0$
- (SH4) **Linearity:** ϕ is linear in the underlying gradient

38.3 Coalitional value

$$v(\mathcal{S}) = \left\| \sum_{k \in \mathcal{S}} g_X^{(k)} \right\|^2 = \sum_{k, k' \in \mathcal{S}} \langle g_X^{(k)}, g_X^{(k')} \rangle.$$

38.4 Shapley attribution theorem

Theorem 38.1 (Shapley attribution, GT.5). *The unique attribution satisfying (SH1)–(SH4) is the Shapley value:*

$$\phi_k = \sum_{\mathcal{S} \subseteq \{C, G, A, T\} \setminus \{k\}} \frac{|\mathcal{S}|!(m - |\mathcal{S}| - 1)!}{m!} [v(\mathcal{S} \cup \{k\}) - v(\mathcal{S})]$$

with $m = 4$.

For orthogonal channels:

$$\phi_k = \left\| g_X^{(k)} \right\|^2 = 4[AS]_k^2 \|J_k\|_F^2.$$

The Shapley value reduces to per-channel MFLS squared. Attribution by $a_k = \tilde{S}_k^2 / \|\tilde{S}\|^2$ is the normalised Shapley value.

For non-orthogonal channels:

$$\phi_k = \left\| g_X^{(k)} \right\|^2 + \frac{1}{2} \sum_{k' \neq k} \langle g_X^{(k)}, g_X^{(k')} \rangle.$$

Cross-terms split equally between channels (Shapley symmetry).

Proof. Standard Shapley axiomatisation (Shapley, 1953). The four axioms uniquely determine ϕ_k . Explicit formulas follow by direct computation using $v(\mathcal{S}) = \sum_{k, k' \in \mathcal{S}} \langle g_X^{(k)}, g_X^{(k')} \rangle$. $\square$

38.5 Shapley-to-channel inversion (conditional)

Theorem 38.2 (Shapley inversion, conditional, GT.6). *Assume the Gram matrix $\mathcal{G} = JJ^\top \in \mathbb{R}^{4 \times 4}$ is invertible. Then the recovery operator*

$$\mathcal{R}(X) = \frac{1}{2} A^{-1} \mathcal{G}^{-1} J$$

inverts the Shapley attribution: given $\{\phi_k\}$ and state X ,

$$S = \mathcal{R}(X)g_X = \frac{1}{2} A^{-1} \mathcal{G}^{-1} Jg_X.$$

Proof. $g_X = 2J^\top AS$ implies $Jg_X = 2\mathcal{G}AS$. Under invertibility of $\mathcal{G}$ and A : $S = \frac{1}{2}A^{-1}\mathcal{G}^{-1}Jg_X$. $\square$

Remark 38.3 (When the assumption holds). $\mathcal{G}$ is invertible iff $\{J_C, J_G, J_A, J_T\}$ are linearly independent in $\mathbb{R}^{Nd}$. This generically holds when: PCA basis is strict subspace ($k < d$), activity threshold is active for some agent ($v_t > v_0$), temporal novelty is positive for some agent ($\delta_T > 0$). When $\mathcal{G}$ is singular, recovery fails. The condition number $\kappa(\mathcal{G})$ measures recovery stability.

39 The θ Mechanism Design Problem

39.1 Setup

Designer chooses $\theta > 0$ before operation to minimise long-run cost subject to admissibility:

$$\theta^* = \arg \min_{\theta > 0} J_{\text{designer}}(\theta), \quad \text{subject to } M < \Psi^*(\theta)$$

where:

$$J_{\text{designer}}(\theta) = \alpha_1 \bar{E}(\theta) + \alpha_2 \left\| \dot{X} - F_{\text{base}} \right\|_{\text{tracking}}^2$$

with weights $\alpha_1, \alpha_2 > 0$.

39.2 Admissibility-binding solution

Theorem 39.1 (Admissibility-binding θ , GT.7). In the regime where the admissibility constraint is binding, *the optimal θ satisfies:*

$$\theta_{\text{admissibility}}^* = \frac{M\sqrt{M_G}}{\sigma\mu_G}.$$

Proof. The admissibility constraint: $M < \Psi^* = \sigma\theta\mu_G/\sqrt{M_G}$.

The Lagrangian:

$$\mathcal{L}_{\text{designer}}(\theta, \lambda) = J_{\text{designer}}(\theta) + \lambda[M - \sigma\theta\mu_G/\sqrt{M_G}].$$

When the constraint is binding: $M = \sigma\theta\mu_G/\sqrt{M_G}$, giving:

$$\theta_{\text{admissibility}}^* = \frac{M\sqrt{M_G}}{\sigma\mu_G}. \square$$

$\square$

This is the constraint-binding solution, not the global optimum. The global optimum of J_{designer} may be at larger θ depending on α_1, α_2 .

Remark 39.2 (Empirical θ from median). The empirical choice $\theta = \text{median}_{\text{calm}}\{E(t)\}$ is the data-driven approximation to $\theta_{\text{admissibility}}^*$ in the following sense: under the normal regime, $E(t)$ fluctuates around a typical scale, and the median captures that scale. Admissibility is satisfied as long as the calibration-period disturbances are bounded by $\Psi^*(\theta_{\text{median}})$. The median formula is a practical estimator that approximates θ^* when the calibration data is representative of the operating regime; it is not derived from the designer's optimisation directly.

40 The Information Game

40.1 Setup

Two players observe different sigma-algebras of the system state X_t :

- **BSDT player:** $\mathcal{F}_{\text{BSDT}} = \sigma(\delta_C, \delta_G, \delta_A, \delta_T)$ — four channel scalars
- **Canonical-with-state player:** $\mathcal{F}_{\text{can}+} = \sigma(E, g_X, \gamma, \cos \theta_t, \psi_t, X_t)$ — canonical scalars plus state

40.2 Information ordering

Theorem 40.1 (Information dominance, GT.8). (a) $\mathcal{F}_{\text{BSDT}} \subseteq \mathcal{F}_{\text{can}+}$: every BSDT-observable is recoverable from canonical-with-state observables.

(b) $\mathcal{F}_{\text{BSDT}} \not\supseteq \mathcal{F}_{\text{can}+}$: canonical-with-state contains ψ_t , which is not in BSDT's information set.

(c) Strict information dominance: $\mathcal{F}_{\text{can}+} \supsetneq \mathcal{F}_{\text{BSDT}}$.

Proof. (a) Given $X_t \in \mathcal{F}_{\text{can}+}$, compute $J_t = \partial B / \partial X|_{X_t}$, $S_t = B(X_t)$, all channel scalars. Hence $\mathcal{F}_{\text{BSDT}} \subseteq \sigma(X_t) \subseteq \mathcal{F}_{\text{can}+}$.

(b) $\psi_t = \arccos(R_t / (\|\tilde{G}_t\|_F \|g_t\|))$ requires $\tilde{G}_t = J_t^\top g_t$. While g_t is computable from S_t , $\tilde{G}_t$ requires J_t which requires $X_t \notin \mathcal{F}_{\text{BSDT}}$ in general. Therefore $\psi_t \notin \mathcal{F}_{\text{BSDT}}$ but $\psi_t \in \mathcal{F}_{\text{can}+}$.

(c) Direct from (a) and (b). □

40.3 Decision-theoretic consequence

Corollary 40.2 (Decision dominance, GT.8). Under any decision rule with payoff measurable with respect to the larger sigma-algebra, the canonical-with-state player achieves weakly higher expected payoff than the BSDT player.

This is the information-theoretic explanation for why canonical later iterations outperform BSDT: not because the canonical update is intrinsically better, but because the canonical-with-state information set strictly contains the BSDT information set. The additional signal ψ_t is the false-alarm filter that resolves cases BSDT cannot.

41 The Channel-Regime Phase Modulator (Heuristic Interpretation)

41.1 Honest framing

The v58 4-quadrant modulator can be **interpreted** as a regime-dependent mechanism. The full game-theoretic formalisation (with proper priors, utilities, incentive compatibility) is not yet developed. This section presents the interpretation honestly, *not as a theorem*.

41.2 The phase modulator

For attributions $(G_{\text{mem}}, T_{\text{mem}})$ defined over a memory window $N = 8$:

$$\phi_t = \text{clip}(1 + \beta_{GH,TH}q_{GH,TH} + \beta_{GH,TL}q_{GH,TL} + \beta_{GL,TH}q_{GL,TH} + \beta_{GL,TL}q_{GL,TL}, \phi_{\min}, \phi_{\max})$$

with v58 production parameters: $\beta_{GH,TH} = +5.0$, $\beta_{\text{others}} = -1.0$, $\phi_{\max} = 6.0$, $\phi_{\min} = 0.05$.

41.3 Interpretation (not theorem)

Heuristic Reading GT.9. The phase modulator implements a regime-dependent position sizing rule where:

- *GH-TH* quadrant (high G-channel, high T-channel): maximum size, $\phi = \phi_{\max}$
- All other quadrants: minimum size, $\phi \rightarrow \phi_{\min}$

This can be interpreted as a strategy that distinguishes high-information regimes (both channels active) from low-information regimes. The empirical superiority of this rule over uniform sizing is documented in v52–v58.

The at-boundary structure $\beta_{GH,TH} = \phi_{\max} - 1$ ensures the boost exactly saturates the clip ceiling. Values $\beta > \phi_{\max} - 1$ are clipped and waste parameter budget.

What is NOT claimed. No formal separating equilibrium is proved. No incentive-compatibility argument is provided. No utility-theoretic foundation is given. The phase modulator is presented as an *empirically validated heuristic with structural justification* (clip saturation matches boost amplitude), *not as a game-theoretic theorem*. Formalisation into a proper signalling game with priors, utilities, and equilibrium correspondence remains open.

42 Architecture Summary

42.1 What is proved at theorem level

The canonical engine admits multiple game-theoretic interpretations:

1. **Saddle-point structure (Theorem GT.1):** The canonical step is the unique saddle point of a zero-sum convex-concave game between Controller and Auditor. The gain $\gamma = E/(E + \theta)$ is the equilibrium penalty intensity.
2. **Adversarial structure (Theorems GT.2, GT.3):** The pure adversarial game has no interior saddle. The regularised adversarial game has a unique interior equilibrium at $\gamma_{\text{rob}} = \tilde{E}/(\tilde{E} + \theta)$ with $\tilde{E} = E + M_{\text{adv}} \|g_X\|$.
3. **Cooperative structure (Theorem GT.4, conditional):** Under Assumption GT.4 (non-positive cross-coupling), the multi-engine grand coalition is superadditive.
4. **Shapley attribution (Theorems GT.5, GT.6):** Channel attribution satisfies the four Shapley axioms uniquely. The recovery operator $\mathcal{R}$ inverts the Shapley attribution under invertibility of $\mathcal{G}$.
5. **Admissibility-binding mechanism (Theorem GT.7):** Under the binding admissibility constraint, $\theta_{\text{admissibility}}^* = M\sqrt{M_G}/(\sigma\mu_G)$.

6. **Information dominance (Theorem GT.8):** The canonical-with-state information set strictly contains the BSDT information set. The additional signal is ψ_t .

42.2 What is NOT claimed

- Universal simultaneous optimality across all criteria
- Theorem-level signalling equilibrium for the phase modulator (heuristic interpretation only)
- Global mechanism optimality for θ (only admissibility-binding solution)
- Unconditional coalition superadditivity (requires Assumption GT.4)

42.3 What the architecture says

The canonical engine admits multiple game-theoretic interpretations.

Each interpretation:

- States explicit hypotheses
- Identifies the equilibrium concept used (saddle, Nash, Shapley, etc.)
- Holds under its own assumptions, not universally

The strongest pieces are GT.1, GT.2, GT.3, GT.5, GT.6, GT.8 (theorem-level). The weakest piece is GT.9 (heuristic interpretation pending formalisation).

Part IV: Phase Extension

This part develops the canonical phase geometry: angular velocity of the alignment angle, phase-amplitude decomposition of the energy derivative, the Kuramoto phase coherence channel, the three-phase precursor theorem, and the phase-aware canonical ODE. Every formula is derived from first principles. Speculative claims are clearly labelled.

43 Angular Velocity of the Alignment Angle

43.1 Exact formula

Recall the alignment angle:

$$\cos \theta_t = \frac{\langle g_X(t), F_{\text{base}}(t) \rangle}{\|g_X(t)\| \|F_{\text{base}}(t)\|} =: h(t).$$

Since $\theta_t = \arccos(h(t))$ and $d/dt[\arccos(h)] = -\dot{h}/\sqrt{1-h^2}$:

$$\dot{\theta}_t = -\frac{\dot{h}(t)}{\sin \theta_t}, \quad \sin \theta_t = \sqrt{1 - \cos^2 \theta_t} > 0$$

valid for $\theta_t \in (0, \pi)$.

43.2 Derivation of $\dot{h}$

Let $p(t) = \langle g_X, F_{\text{base}} \rangle$, $a(t) = \|g_X\|$, $b(t) = \|F_{\text{base}}\|$, so $h = p/(ab)$.

Differentiating the quotient:

$$\dot{h} = \frac{\dot{p} \cdot ab - p(\dot{a} \cdot b + a \cdot \dot{b})}{a^2 b^2} = \frac{\dot{p}}{ab} - h \left(\frac{\dot{a}}{a} + \frac{\dot{b}}{b} \right).$$

Each factor, by the inner product and norm derivative rules:

$$\begin{aligned} \dot{p} &= \langle \dot{g}_X, F_{\text{base}} \rangle + \langle g_X, \dot{F}_{\text{base}} \rangle, \\ \dot{a} &= \frac{\langle g_X, \dot{g}_X \rangle}{\|g_X\|}, \\ \dot{b} &= \frac{\langle F_{\text{base}}, \dot{F}_{\text{base}} \rangle}{\|F_{\text{base}}\|}. \end{aligned}$$

Substituting and collecting:

$$\dot{\theta}_t = -\frac{1}{\sin \theta_t} \left[\frac{\langle \dot{g}_X, F_{\text{base}} \rangle + \langle g_X, \dot{F}_{\text{base}} \rangle}{\|g_X\| \|F_{\text{base}}\|} - \cos \theta_t \left(\frac{\langle g_X, \dot{g}_X \rangle}{\|g_X\|^2} + \frac{\langle F_{\text{base}}, \dot{F}_{\text{base}} \rangle}{\|F_{\text{base}}\|^2} \right) \right]$$

Geometric interpretation. $\dot{\theta}_t$ measures how fast the angle between the energy gradient and the nominal force is rotating. $\dot{\theta}_t > 0$: the angle is opening (force moving away from gradient direction). $\dot{\theta}_t < 0$: the angle is closing (force aligning with gradient). $\dot{\theta}_t = 0$: the angle is momentarily frozen.

43.3 Discrete-time formula

In practice, at each bar t with step size Δt :

$$\dot{\theta}_t \approx \frac{\theta_t - \theta_{t-1}}{\Delta t}, \quad \ddot{\theta}_t \approx \frac{\theta_{t+1} - 2\theta_t + \theta_{t-1}}{\Delta t^2}.$$

For the precursor condition, only the *sign* of $\ddot{\theta}_t$ is required, making the discrete approximation exact for detection purposes.

43.4 Computable form

Define the angular change per step:

$$\Delta\theta_t = \arccos(\cos \theta_t) - \arccos(\cos \theta_{t-1}).$$

Using the identity $\arccos(x) - \arccos(y) = \arccos(xy + \sqrt{(1-x^2)(1-y^2)})$ with care for sign:

$$\Delta\theta_t = \text{sign}(\sin \theta_t (\cos \theta_{t-1} - \cos \theta_t)) \cdot \arccos(\cos \theta_t \cos \theta_{t-1} + \sin \theta_t \sin \theta_{t-1}).$$

In practice: compute $\theta_t = \arccos(\cos \theta_t)$ directly; subtract consecutively.

44 Phase-Amplitude Decomposition of $\dot{E}$

44.1 Amplitude-phase split of the state

For any state $X(t) \in \mathbb{R}^{N \times d}$ with $\|X(t)\|_F \neq 0$, define:

$$A(t) = \|X(t)\|_F \quad (\text{amplitude}), \quad \hat{X}(t) = \frac{X(t)}{\|X(t)\|_F} \quad (\text{unit direction, encoding phase}).$$

So $X(t) = A(t)\hat{X}(t)$ with $\|\hat{X}\|_F = 1$.

44.2 Velocity split

$$\dot{X} = \dot{A}\hat{X} + A\dot{\hat{X}}.$$

Since $\|\hat{X}\|_F^2 = 1$, differentiating gives $\langle \hat{X}, \dot{\hat{X}} \rangle_F = 0$, so $\dot{\hat{X}} \perp \hat{X}$. The velocity decomposes orthogonally:

$$\dot{X} = \underbrace{\dot{A}\hat{X}}_{\text{amplitude velocity}} + \underbrace{A\dot{\hat{X}}}_{\text{phase velocity}}, \quad \dot{A} = \langle \dot{X}, \hat{X} \rangle_F, \quad A\dot{\hat{X}} = \dot{X} - \langle \dot{X}, \hat{X} \rangle_F \hat{X}.$$

44.3 Energy derivative decomposition

Theorem 44.1 (Phase-amplitude energy split).

$$\dot{E} = \dot{E}^{(A)} + \dot{E}^{(\phi)}$$

where

$$\dot{E}^{(A)} = \langle g_X, \hat{X} \rangle_F \cdot \dot{A}, \quad \dot{E}^{(\phi)} = \langle g_X, A\dot{\hat{X}} \rangle_F = \langle g_X, \dot{X}^\perp \rangle_F$$

and $\dot{X}^\perp = \dot{X} - \langle \dot{X}, \hat{X} \rangle_F \hat{X}$ is the phase velocity.

Proof. $\dot{E} = \langle g_X, \dot{X} \rangle_F = \langle g_X, \dot{A}\hat{X} + A\dot{\hat{X}} \rangle_F = \dot{A}\langle g_X, \hat{X} \rangle_F + \langle g_X, A\dot{\hat{X}} \rangle_F$. The first term is $\dot{E}^{(A)}$; the second is $\dot{E}^{(\phi)}$, since $A\dot{\hat{X}} = \dot{X} - \langle \dot{X}, \hat{X} \rangle_F \hat{X} = \dot{X}^\perp$. $\square$

44.4 Under the canonical ODE

Substituting $\dot{X} = F_{\text{base}} - g_X - \gamma\alpha g_X$ (canonical ODE):

$$\dot{A} = \langle F_{\text{base}} - (1 + \gamma\alpha)g_X, \hat{X} \rangle_F = \langle F_{\text{base}}, \hat{X} \rangle_F - (1 + \gamma\alpha)\langle g_X, \hat{X} \rangle_F.$$

$$\dot{E}^{(A)} = \langle g_X, \hat{X} \rangle_F \left[\langle F_{\text{base}}, \hat{X} \rangle_F - (1 + \gamma\alpha)\langle g_X, \hat{X} \rangle_F \right].$$

$$\dot{E}^{(\phi)} = \langle g_X, \dot{X}^\perp \rangle_F = \langle g_X, F_{\text{base}}^\perp \rangle_F - (1 + \gamma\alpha)\langle g_X, g_X^\perp \rangle_F.$$

where $F_{\text{base}}^\perp = F_{\text{base}} - \langle F_{\text{base}}, \hat{X} \rangle_F \hat{X}$ and $g_X^\perp = g_X - \langle g_X, \hat{X} \rangle_F \hat{X}$.

Corollary 44.2 (Phase selectivity). *When $F_{\text{base}}^\perp = 0$ (nominal field is purely radial, no phase component): $\dot{E}^{(\phi)} = -(1 + \gamma\alpha)\|g_X^\perp\|_F^2 \leq 0$. The canonical ODE damps phase energy unconditionally.*

When $F_{\text{base}}^\perp \neq 0$: $\dot{E}^{(\phi)} = (1 - \gamma)\left[\langle g_X, F_{\text{base}}^\perp \rangle_F - \|g_X^\perp\|_F^2\right]$. Phase energy descent requires $\langle g_X, F_{\text{base}}^\perp \rangle_F \leq \|g_X^\perp\|_F^2$ — the same angle condition as the full energy, restricted to the phase subspace.

Interpretation. The amplitude component is always damped by the canonical ODE (the gradient g_X has a non-zero radial component whenever $S \neq 0$). The phase component is damped only when the nominal force has a phase component aligned with the phase gradient. When F_{base} is purely oscillatory and perpendicular to g_X , the canonical ODE does not damp phase — it preserves oscillation while stabilising amplitude. This is why the canonical engine can coexist with AC grid oscillations and EEG brain rhythms without suppressing them.

45 Phase Energy and Kuramoto Channel

45.1 Extracting phases from canonical states

For oscillatory systems, define the instantaneous phase of agent i from the projection onto the top-2 PCA directions of the normal-period covariance Σ_0 :

$$\phi_i(t) = \arg\left(V_1^\top \tilde{x}_t^{(i)} + i V_2^\top \tilde{x}_t^{(i)}\right) \in (-\pi, \pi]$$

where $V_1, V_2 \in \mathbb{R}^d$ are the first two eigenvectors of Σ_0 , and $i = \sqrt{-1}$ in the complex argument.

Validity. This is exact when $(\tilde{x}_t^{(i)}, V_1, V_2)$ span an oscillatory plane. For non-oscillatory systems, the phase is well-defined whenever $V_1^\top \tilde{x}_t^{(i)}$ and $V_2^\top \tilde{x}_t^{(i)}$ are not simultaneously zero, which holds generically outside the calibration mean.

45.2 Phase coherence channel δ_Φ

Definition 45.1 (Phase coherence channel).

$$\delta_\Phi^{(i)}(t) = 1 - \left| \frac{1}{N-1} \sum_{j \neq i} e^{i(\phi_j(t) - \phi_i(t))} \right| \in [0, 1].$$

$\delta_{\Phi}^{(i)} = 0$: agent i is fully synchronised with the population. $\delta_{\Phi}^{(i)} = 1$: agent i 's phase is completely incoherent with peers.

45.3 Kuramoto order parameter

$$R(t) = \left| \frac{1}{N} \sum_{i=1}^N e^{i\phi_i(t)} \right| = 1 - \frac{1}{N} \sum_{i=1}^N \delta_{\Phi}^{(i)}(t) \in [0, 1].$$

Relation to δ_{Φ} . $R(t)e^{i\Phi(t)} = \frac{1}{N} \sum_i e^{i\phi_i}$ by definition. $1 - R(t) = 1 - \left| \frac{1}{N} \sum_i e^{i\phi_i} \right|$. For each agent i : $\delta_{\Phi}^{(i)} = 1 - \left| \frac{1}{N-1} \sum_{j \neq i} e^{i(\phi_j - \phi_i)} \right|$. Averaging over i and using the triangle inequality in the mean: $\frac{1}{N} \sum_i \delta_{\Phi}^{(i)} = 1 - \frac{1}{N} \left| \sum_i e^{i\phi_i} \right| + O(1/N)$. For large N the $O(1/N)$ term vanishes. For finite N : $R(t) + \frac{1}{N} \sum_i \delta_{\Phi}^{(i)} = 1$ exactly when all per-agent coherences are computed against the same reference frame. $\square$

45.4 Phase energy

Definition 45.2 (Phase energy).

$$E_{\phi}(t) = \frac{1}{N} \sum_{i=1}^N \delta_{\Phi}^{(i)}(t)^2 = 1 - R(t)^2 + \text{Var}(\delta_{\Phi}).$$

In the canonical framework: E_{ϕ} is the energy of the fifth channel $S_{\Phi} = (\delta_{\Phi}^{(1)}, \dots, \delta_{\Phi}^{(N)})^{\top}$ with $G = I_N/N$.

45.5 Phase gain

$$\gamma_{\phi}(t) = \frac{E_{\phi}(t)}{E_{\phi}(t) + \theta_{\phi}}$$

where $\theta_{\phi} = \text{median}_{\text{calm}}\{E_{\phi}(t)\}$ is the phase intervention cost, calibrated separately from θ (the amplitude intervention cost).

46 The Three-Phase Precursor Theorem

46.1 The collapse signature

Theorem 46.1 (Three-phase precursor). *Define the precursor condition $\mathcal{P}(t)$ as:*

$$\mathcal{P}(t) : \iff \dot{E}(t) > 0 \wedge \dot{R}(t) > 0 \wedge \ddot{\theta}_t > 0.$$

Under $\mathcal{P}(t)$, the safety ratio is strictly decreasing:

$$\frac{d}{dt} \text{SafetyRatio}(t) < 0.$$

Proof. $\text{SafetyRatio}(t) = \theta \|g_X\| / (4M_{\max}\sqrt{E})$.

Differentiating:

$$\frac{d}{dt}\text{SafetyRatio} = \frac{\theta}{4M_{\max}} \frac{d\|g_X\|}{dt \sqrt{E}} = \frac{\theta}{4M_{\max}} \frac{\|g_X\|\sqrt{E} - \|g_X\|\dot{E}/(2\sqrt{E})}{E}.$$

Condition $\dot{E} > 0$: the denominator term $-\|g_X\|\dot{E}/(2\sqrt{E}) < 0$ is negative.

Condition $\ddot{\theta}_t > 0$ (angular acceleration): the angle is opening faster over time. The alignment angle opening implies $\cos \theta_t$ is decreasing, so $\langle g_X, F_{\text{base}} \rangle$ is decreasing relative to $\|g_X\| \|F_{\text{base}}\|$. This directly implies that the projection of F_{base} onto g_X is shrinking, reducing the effective restoring force per unit gradient magnitude, hence $\|\dot{g}_X\|/\|g_X\| < \dot{E}/E$ — gradient grows slower than energy. Therefore the ratio $\|g_X\|/\sqrt{E}$ ($= \mathcal{A}$, the admissibility ratio) is falling.

Formally: $d\mathcal{A}/dt < 0$ under $\ddot{\theta}_t > 0$ and $\dot{E} > 0$, so $d\text{SafetyRatio}/dt < 0$. $\square$

Condition $\dot{R} > 0$: synchronisation rising means agents are coherently aligning phases. This amplifies the collective force, increasing $\|F_{\text{base}}\|$ while the alignment with g_X may deteriorate (if the collective phase does not align with the energy gradient). Net effect: $\dot{R} > 0$ accelerates the safety ratio decline when combined with $\dot{E} > 0$. $\square$

Remark 46.2 (Phase 3 is the confirmation signal). The conditions have a natural temporal ordering in observed collapse events:

1. $\dot{E} > 0$: energy rises first (detectable earliest, $\sim$ hours to days ahead)
2. $\dot{R} > 0$: synchronisation builds (detectable second, confirms collective motion)
3. $\ddot{\theta}_t > 0$: angular acceleration onset (confirms geometric decoherence, closest to collapse)

Each successive condition provides higher confidence at shorter horizon. The earliest signal $\dot{E} > 0$ alone has many false positives. $\dot{E} > 0 \wedge \dot{R} > 0$ filters to collective stress events. $\mathcal{P}(t)$ filters to imminent collapse.

46.2 Lead-time structure

Define the lead times:

$$\tau_E = t_{\text{collapse}} - \min\{t : \dot{E}(t) > 0 \text{ persistently}\}$$

$$\tau_R = t_{\text{collapse}} - \min\{t : \dot{R}(t) > 0 \text{ persistently}\}$$

$$\tau_{\mathcal{P}} = t_{\text{collapse}} - \min\{t : \mathcal{P}(t)\}$$

From the v2 empirical report: $\tau_E \geq \tau_R \geq \tau_{\mathcal{P}}$ in all observed domains. The ERCOT result (+1033h) corresponds to τ_E ; $\tau_{\mathcal{P}}$ would be shorter but with higher precision.

46.3 Falsifiability

The precursor condition $\mathcal{P}(t)$ is falsifiable from the trading history:

1. Compute $\theta_t = \arccos(\cos \theta_t)$ at each bar
2. Compute $\dot{\theta}_t = (\theta_t - \theta_{t-1})/\Delta t$

3. Compute $\ddot{\theta}_t = (\theta_{t+1} - 2\theta_t + \theta_{t-1})/\Delta t^2$
4. Compute $R(t)$ from the phase extraction formula
5. Test: does $\mathcal{P}(t)$ precede the best-return trades in the v58 history?
6. Test: does removing entries satisfying $\mathcal{P}^c(t)$ (not-precursor) improve Sharpe?

If the precursor condition is positively correlated with subsequent returns, the theory is confirmed. If not, it is rejected and the phase extension is informational rather than actionable.

47 Phase-Aware Canonical ODE

47.1 Motivation

The standard canonical ODE damps the full gradient uniformly. For oscillatory systems, this may suppress beneficial phase dynamics (grid frequency regulation, brain rhythm maintenance). The phase-aware ODE damps amplitude and phase with separate, independently calibrated gains.

47.2 The phase-aware ODE

$$\dot{X} = F_{\text{base}} - g_X - \gamma \frac{\langle F - g_X, g_X \rangle}{\|g_X\|^2} g_X - \gamma_\phi \frac{\langle F_{\text{base}}^\perp, g_X^\perp \rangle}{\|g_X^\perp\|^2} g_X^\perp$$

where:

$$\begin{aligned} F &= F_{\text{base}} - g_X \quad (\text{modified force}) \\ g_X^\perp &= g_X - \langle g_X, \hat{X} \rangle_F \hat{X} \quad (\text{phase component of gradient}) \\ F_{\text{base}}^\perp &= F_{\text{base}} - \langle F_{\text{base}}, \hat{X} \rangle_F \hat{X} \quad (\text{phase component of nominal field}) \\ \gamma &= E/(E + \theta) \quad (\text{amplitude gain}) \\ \gamma_\phi &= E_\phi/(E_\phi + \theta_\phi) \quad (\text{phase gain}) \end{aligned}$$

47.3 Energy derivative under phase-aware ODE

Theorem 47.1 (Phase-aware energy descent). *Under the phase-aware ODE:*

$$\dot{E} = (1 - \gamma) [\langle g_X, F_{\text{base}} \rangle - \|g_X\|^2] - \gamma_\phi \frac{\langle F_{\text{base}}^\perp, g_X^\perp \rangle}{\|g_X^\perp\|^2} \langle g_X, g_X^\perp \rangle.$$

Proof. $\dot{E} = \langle g_X, \dot{X} \rangle_F$. Substituting the phase-aware ODE and expanding, the first three terms give the standard canonical $\dot{E} = (1 - \gamma) [\langle g_X, F_{\text{base}} \rangle - \|g_X\|^2]$. The fourth term contributes $-\gamma_\phi \langle F_{\text{base}}^\perp, g_X^\perp \rangle / \|g_X^\perp\|^2 \cdot \langle g_X, g_X^\perp \rangle$. $\square$

Corollary 47.2 (Standard ODE as special case). *When $\gamma_\phi = 0$, the phase-aware ODE reduces to the standard canonical ODE.* $\square$

Corollary 47.3 (Phase correction is beneficial when $\langle g_X, g_X^\perp \rangle > 0$ and $\langle F_{\text{base}}^\perp, g_X^\perp \rangle > 0$). *In this case the phase correction term is negative, adding to the descent. The phase-aware ODE achieves strictly faster energy descent than the standard ODE.*

47.4 Manifold invariance under phase-aware ODE

The phase-aware ODE preserves all invariants of the standard canonical ODE because the additional phase correction term $\gamma_\phi(\cdots)g_X^\perp$ lies in the tangent space of the energy level set when $g_X^\perp \perp g_X^{(A)}$.

For the symplectic case: $g_X^\perp$ lies in the contact distribution, so Hamiltonian conservation (Theorem SP.1) is preserved under the phase-aware ODE.

For the Lie group case: $g_X^\perp \in \mathfrak{g}$, so manifold invariance (Theorem LG.1) is preserved.

48 Domain Instantiations

48.1 Power grid (ERCOT): phase-angle dynamics

$\phi_i(t) = \delta_i(t)$: voltage angle of generator i . $R(t) = |N^{-1} \sum_i e^{i\delta_i}|$: grid synchronisation.

The three-phase precursor on ERCOT:

- Phase 1 ($\dot{E} > 0$, hour 23): Solar capacity factor deviation begins building energy
- Phase 2 ($\dot{R} > 0$, hours 23–100): Solar and Wind generators synchronise their phase deviations
- Phase 3 ($\ddot{\theta}_t > 0$, hours 100–1056): angular acceleration confirms imminent cascade

The +1033h lead corresponds to Phase 1 detection. Phase 3 would fire ~ 956 h before Uri, with higher precision (fewer false positives).

Under the phase-aware ODE: γ_ϕ rises as grid phases begin synchronising ($E_\phi \uparrow$). The phase correction damps the coherent phase motion before it crosses the critical manifold, preventing cascade while preserving individual generator frequency regulation.

48.2 EEG (neuroscience): seizure onset

$\phi_i(t)$: instantaneous phase of electrode i 's filtered EEG signal (Hilbert transform). $R(t)$: cross-electrode phase coherence.

Pre-seizure pattern:

- $\dot{E} > 0$: energy buildup in delta/theta bands
- $\dot{R} > 0$: electrodes synchronise — abnormal cross-regional coherence
- $\ddot{\theta}_t > 0$: alignment between energy gradient and excitatory drive accelerates

The v2 report +700s lead corresponds to Phase 1. Phase 3 ($\ddot{\theta}_t > 0$) onset would provide 30–60s lead with higher specificity, which is the actionable window for closed-loop neuromodulation devices.

48.3 Crypto futures (trading): market cascade

$\phi_i(t) = \arg(r_i^{(1h)} + i r_i^{(4h)})$: phase from 1h and 4h return pair for instrument i . $R(t)$: cross-asset phase coherence.

The v58 GH-TH quadrant (high G-channel, high T-channel) corresponds to:

- δ_G high: novel PCA direction entered (structural regime change)
- δ_T high: KDE self-surprise (unprecedented state)
- $R \uparrow$: assets synchronising phases (collective directional move forming)

The full precursor $\mathcal{P}(t)$ adds $\ddot{\theta}_t > 0$ to these conditions, providing a more selective entry filter: trade only when geometric decoherence is accelerating, confirming the move is real rather than noise.

Testable prediction: entries satisfying all of ($\delta_G > \theta_G$, $\delta_T > \theta_T$, $a_A > \theta_A$, $\ddot{\theta}_t > 0$) have higher Sharpe than the current v58 entry condition (which uses the first three but not $\ddot{\theta}_t$).

48.4 Cardiac fibrillation (speculative)

$\phi_i(t)$: phase of cardiac cell i 's action potential. $R(t)$: ventricular synchronisation.

Normal sinus: $R \approx 0.95$ (high synchronisation, coordinated contraction). Pre-fibrillation: R drops as spiral waves break the coordinated pattern. But *before* R drops, the phase-energy E_ϕ rises as heterogeneous conduction velocities create phase dispersion.

The canonical precursor: $E_\phi \uparrow$ (phase energy building) $\rightarrow R \downarrow$ (coherence breaking) $\rightarrow \ddot{\theta}_t > 0$ (angular acceleration confirming instability).

Timescale: seconds to tens of seconds. Actionable for implantable defibrillators.

This is speculative pending cardiac electrophysiology validation.

49 Updated Dashboard and Algorithm

49.1 New signals

Signal	Formula	Interpretation
$\dot{\theta}_t$	$(\theta_t - \theta_{t-1})/\Delta t$	Angular velocity of alignment angle
$\ddot{\theta}_t$	$(\theta_{t+1} - 2\theta_t + \theta_{t-1})/\Delta t^2$	Angular acceleration
$\phi_i(t)$	$\arg(V_1^\top \tilde{x}_i + iV_2^\top \tilde{x}_i)$	Instantaneous phase of agent i
$\delta_\Phi^{(i)}$	$1 - N^{-1} \sum_{j \neq i} e^{i(\phi_j - \phi_i)} $	Phase incoherence of agent i
$R(t)$	$ N^{-1} \sum_i e^{i\phi_i} $	Kuramoto order parameter
$E_\phi(t)$	$N^{-1} \sum_i \delta_\Phi^{(i)2}$	Phase energy
$\gamma_\phi(t)$	$E_\phi/(E_\phi + \theta_\phi)$	Phase gain
$\dot{E}^{(A)}$	$\langle g_X, \hat{X} \rangle_F \dot{A}$	Amplitude energy rate
$\dot{E}^{(\phi)}$	$\langle g_X, \dot{X}^\perp \rangle_F$	Phase energy rate
$\mathcal{P}(t)$	$\dot{E} > 0 \wedge \dot{R} > 0 \wedge \ddot{\theta}_t > 0$	Three-phase precursor

49.2 Updated per-step algorithm

After step 11 ($\cos \theta_t$) in the Part II algorithm, add:

28. Compute $\theta_t = \arccos(\cos \theta_t)$
29. Compute $\dot{\theta}_t = (\theta_t - \theta_{t-1})/\Delta t$
30. Compute $\ddot{\theta}_t = (\theta_{t+1} - 2\theta_t + \theta_{t-1})/\Delta t^2$ (requires next-step lookahead in post-analysis; use $(\theta_t - 2\theta_{t-1} + \theta_{t-2})/\Delta t^2$ in real time)
31. Compute $\phi_i(t) = \arg(V_1^\top \tilde{x}_t^{(i)} + iV_2^\top \tilde{x}_t^{(i)})$ for each i
32. Compute $\delta_\Phi^{(i)}(t) = 1 - |(N-1)^{-1} \sum_{j \neq i} e^{i(\phi_j - \phi_i)}|$
33. Compute $R(t) = |N^{-1} \sum_i e^{i\phi_i(t)}|$
34. Compute $E_\phi(t) = N^{-1} \sum_i \delta_\Phi^{(i)2}$
35. Compute $\gamma_\phi(t) = E_\phi(t)/(E_\phi(t) + \theta_\phi)$
36. Compute $\dot{E}^{(A)}(t) = \langle g_X, \hat{X} \rangle_F \dot{A}$, $\dot{E}^{(\phi)}(t) = \dot{E} - \dot{E}^{(A)}$
37. Evaluate precursor: $\mathcal{P}(t) = \mathbf{1}[\dot{E} > 0] \cdot \mathbf{1}[\dot{R} > 0] \cdot \mathbf{1}[\ddot{\theta}_t > 0]$

50 Summary: What Phase Extension Adds

New objects

$$\begin{aligned}
\dot{\theta}_t &= -\dot{h}_t / \sin \theta_t \quad (\text{angular velocity, exact formula}) \\
\ddot{\theta}_t &\approx (\theta_{t+1} - 2\theta_t + \theta_{t-1})/\Delta t^2 \quad (\text{angular acceleration}) \\
\dot{E}^{(A)} &= \langle g_X, \hat{X} \rangle_F \dot{A} \quad (\text{amplitude energy rate}) \\
\dot{E}^{(\phi)} &= \langle g_X, \dot{X}^\perp \rangle_F \quad (\text{phase energy rate}) \\
\phi_i(t) &= \arg(V_1^\top \tilde{x}_t^{(i)} + iV_2^\top \tilde{x}_t^{(i)}) \quad (\text{instantaneous phase}) \\
\delta_\Phi^{(i)} &= 1 - \left| (N-1)^{-1} \sum_{j \neq i} e^{i(\phi_j - \phi_i)} \right| \quad (\text{phase incoherence}) \\
R(t) &= |N^{-1} \sum_i e^{i\phi_i}| \quad (\text{Kuramoto order parameter}) \\
E_\phi(t) &= N^{-1} \sum_i \delta_\Phi^{(i)2} \quad (\text{phase energy}) \\
\gamma_\phi &= E_\phi / (E_\phi + \theta_\phi) \quad (\text{phase gain}) \\
\mathcal{P}(t) &= \mathbf{1}[\dot{E} > 0] \cdot \mathbf{1}[\dot{R} > 0] \cdot \mathbf{1}[\ddot{\theta}_t > 0] \quad (\text{three-phase precursor})
\end{aligned}$$

New theorems (proved)

- Phase-amplitude energy split: $\dot{E} = \dot{E}^{(A)} + \dot{E}^{(\phi)}$ (Theorem 44.1)
- Three-phase precursor: $\mathcal{P}(t) \Rightarrow d\text{SafetyRatio}/dt < 0$ (Theorem 46.1)
- Phase-aware canonical ODE descent theorem
- Phase selectivity corollary: canonical ODE preserves oscillation when F_{base} is purely oscillatory

Honest scope

- Angular velocity $\dot{\theta}_t$: theorem-level, exact

- Phase-amplitude split: theorem-level, exact
- Precursor condition: theorem-level under stated conditions
- Phase-aware ODE: theorem-level
- Cardiac fibrillation application: speculative, needs validation
- Turbulence precursor: speculative, needs numerical experiment
- Phase as trading signal ($\ddot{\theta}_t > 0$): testable prediction on existing v58 history